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NLP for education

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Extended Abstract

Natural Language Processing (NLP) is a form of Artificial Intelligence. It programs human language to a machine-readable format to achieve human-computer direct interaction. The importance of NLP-based virtual course assistants has raised its reputation to perform different education-related purposes. This paper proposes to develop an NLP-based virtual course assistant to provide educational support for both teachers and students. We believe adopting NLP-based virtual course assistant in class bring improvements in both teaching and learning experiences. With the help of an NLP-based virtual course assistant, it answers general and straightforward questions about the course 24/7. For example, students will ask the teacher many common questions, such as the weighting of each coursework, the time and place of the first lesson, etc. Yet, with a limited number of teaching assistants and teachers, answering similar questions to many students is time-consuming. It would be beneficial if the virtual course assistants could answer general questions about the course. So, the teaching team could devote more effort and time to answering complex questions. For students, an immediate response increases their motivation to study. It also avoids the embarrassment of directly approaching the teacher. The paper reviews four kinds of NLP tasks which are 1) Dialogue, 2) Question Answering (QA), 3) Question Generation, and 4) Sentiment Analysis (SA). Furthermore, the paper examines the performance of approaches to accomplish NLP tasks, including 1) Rule-based, 2) Chatbot Builder, and 3) Deep Learning.

The project scope defines as constructing a virtual course assistant for university students and teachers. The design of the NLP-based virtual course assistant needs to consider two main components: 1) Data and 2) NLP tasks. Firstly, there are no suitable datasets for the task proposed in this project. We need to gather different kinds of course data. Also, a sustainable method for further data collection is to develop a web-based course registration system for users to add and update course information. Appropriate access control and data integrity will need to concern while developing the web application. Secondly, to meet the need of teachers and students, a virtual course assistant will need to perform two kinds of NLP tasks, including QA, and SA. We adopt Bidirectional Encoder Representations from Transformers (BERT) as the key structure of models. We will fine-tune the models to meet different purposes.

The paper mentioned the performance evaluation design of the web-based course registration system and each NLP task performed by a virtual course assistant. To begin with, we will evaluate the web application according to a set of Website Performance Measurement metrics defined by Ghattas et al. (2020). Then, as some of our NLP tasks are closed domain, we will create our testing

data to test the model and get the classification accuracy score. For the open domain SA task, we will test the model on the 100k-courseras-course-reviews-dataset to evaluate the model performance.

Acknowledgement

Dr. Song Linqi, my final year project supervisor, deserves my heartfelt appreciation. Dr. Song set up meetings with me on a regular basis to examine my progress and see if there is anything he can do to help. He always provides me with helpful feedback on my completed components. He also went out of his way to assist me in putting together a data collection. I would also want to express my gratitude to Ms. Liu Shuqi for her guidance and support in keeping my work on track. She was always willing to meet with me and provided me with a wealth of learning resources to help me learn about natural language processing. My final year assignment could not be done smoothly without both of them patience in guiding and active involvement. It is an honor for me to have Dr. Song and Ms. Liu supervise my project.

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1. Introduction

1.1 Background

The most important goal for schools is to attain high student achievement, which means that students understand the majority of academic information in a particular time frame. Minimizing the student-teacher ratio has been proven to be one of the determinants of student achievement in previous studies. Koc and Celik (2005) discovered a negative correlation between the average number of students per teacher and exam ranking of cities. It indicates that the lower number of students per teacher tends to have a higher ranking on the exam. Chapman et al. (2010) also found that larger class sizes are associated with a poor perception of student learning. With this in mind, most countries tend to have a class reduction, decreasing class size to provide a better learning experience for students (Blatchford et al., 2011).

However, the insufficient teaching staff is a common stumbling block to class reduction. Worse still, as the COVID-19 pandemic spreads, online education is gaining more prominence because schools, colleges, and universities shut down for an indefinite time (Martinez, 2020). Since the dawn of online teaching, students may require additional support from teaching teams. If the class size is large, communication and the needs of the students will be significantly increased. However, some of the questions posed by students may be similar and straightforward (Goel et al.,2016). Answering these questions repeatedly through different channels, such as email and online discussion boards, is time-consuming. Students might constantly send emails to teachers for follow-up replies as the communication is not efficient like face-to-face. (Sepulveda-Escobar et al., 2020). Under this circumstance, the teaching team will stress with a high and redundant workload.

A viable solution to the above difficulties is involving a virtual assistant in class. Virtual Assistant (VA) provides corresponding responses and services swiftly through conversation between user and machine (Kincaid et al., 2017). It indicates that VA could share the teaching team's work, such as answering simple questions about courses and assignments 24/7. Thus, students can get immediate responses and avoid the embarrassment of directly approaching the teacher. Also, the teaching team could devote effort and time to answer complex questions and participate in deeper discussions with students. The implementation of VA's conversational access is achieved by a critical technology called Natural Language Progressing (NLP). NLP is a branch of artificial intelligence that enables machines to extract meaning from human languages to establish direct

human-computer interaction (Kalburgikar et al., 2021). Regarding VA's remarkable features, it is fast becoming a key instrument in advancement in education.

1.2 Problem Definition and Scope

This project will investigate a solution to develop an all-time applicable virtual course assistant that can provide educational support to teachers and students. Even if course information is constantly changing, the virtual course assistant should function successfully throughout each academic year. Database and NLP models are two key components that must be examined in order to achieve the goal.

Firstly, since course information may frequently change in response to teacher requests or for any other academic purpose, the database would need to be updated on a regular basis. As a result, responsible users should be able to add, update, and alter course information. While considering the approach, data integrity and appropriate access control would need to be controlled.

Secondly, several NLP models will be developed to fulfill the educational needs of both teachers and students. Multi class classification, question answering, and sentiment analysis are among the NLP tasks that will be done.

Regarding the above needs, the scope will be defined to construct a virtual course assistant for university students and teachers. Course virtual assistant also integrates with a web-based course information system, allowing responsible users to update course information.

The project will achieve five specific goals.

1. The web-based course registration system is well designed to maintain data integrity and proper access control.
2. The NLP-based virtual course assistant understands natural language and responds with valuable answers swiftly.
3. The NLP-based virtual course assistant provides functions to assist students in confirming and improving their level of understanding about courses.
4. The NLP-based virtual course assistant performs sentiment analysis according to student comments.
5. The NLP-based virtual course assistant creates statistical data regarding users' inquiries and performances to provide feedback to educators.

1.3 Problem Significance and Motivation

The implication of the introduction of virtual course assistants to class is massive and has the possibility to strengthen individuals' learning outcomes. It provides improvements and hopefully brings new perspectives to the development of virtual educational assistants, which will be discussed in the literature study.

1.4 Paper Structure

According to the above descriptions, we propose to develop an NLP-based virtual course assistant which facilitates question-answering and quality assessment of multi-courses. The capabilities of NLP-based virtual course assistants will be further expanded by adding user feedback and emotional support features. The rest of this paper is organized as follows: Section 2 reviews related works and indicates the necessity of our work. Section 3 contains preliminary solution descriptions and diagrams of our system architecture. The methodology and implications are briefly explained in section 4. A conclusion of our work and future improvement are drawn in section 5.

2. Literature Review

The proposed virtual course assistants would need to perform four kinds of NLP tasks, including Dialogue, Question Answering, Question Generation, Sentiment Analysis. There are different techniques to complete some of the above functions, while their performance may vary. Thus, it is necessary to understand the nature of those NLP tasks. Then, review different approaches and evaluate their capabilities and limitations. Last but not least, existing educational virtual assistants constructed will also be examined.

2.1 Overview of NLP tasks

2.1.1 Dialogue

Dialogue is an NLP task commonly seen in real-life applications, like Chatbot. The human could directly interact with machines via conversation. Ni et al. (2021) explained the type of dialogue that is task-oriented and open-domain. The task-oriented dialogue will only interact with users in a certain domain, such as movie ticket booking and restaurant table reserving. Open-domain dialogue allows users to chat with no domain restrictions. The dialogue tasks would be performed with other NLP tasks simultaneously.

2.1.2 Question Answering (QA)

Bouziane et al. (2015) defined QA as answering questions written in natural language automatically. They also identified three significant components of QA: Question Analysis, Document Retrieval, and Answer Extraction. The QA generally adheres to these three subtasks (Chiu et al.,2020; Eicher et al.,2018; Harvey et al., 2015).

2.1.3 Question Generation (QG)

Question generation (QG) comprehends a given text and generates related questions in natural language. QG is commonly applied to education. Mitkov et al. (2003) prepared exam questions for a linguistics course in a semi-automated way. This method is more cost-efficient, and the questions are of comparable quality to those created manually. Hoshino et al. (2005) created a real-time system that uses web news articles to generate English grammar and vocabulary questions. Du et al. (2017) proposed a fully data-driven neural network approach to generate reading comprehension questions automatically.

2.1.4 Sentiment Analysis (SA)

Sentiment analysis (SA) derives the emotion from a given text. There are three types of sentiment analysis approaches: lexicon-based methods, machine learning-based methods, and hybrid methods. Mite-Baidal et al. (2018) disclosed that employing sentiment analysis in education is common and improves student performance, teaching process, and course satisfaction. They also found that Facebook and Twitter are the commonly used social network dataset for sentiment analysis in the education domain.

2.2 Overview of Current Approaches to Perform NLP Tasks

2.2.1 Rule-based Approach

In performing dialogue and QA tasks, the rule-based approach uses an algorithm that processes user utterances through a prepared set of rules (Singh et al.,2019). Weizenbaum (1996) used rule-based orientation to develop ELIZA, the first computer program for natural language processing between man and machine. When an input sentence is provided, ELIZA could recognize keywords and match those keywords against a set of pre-defined criteria to generate suitable responses. The method employed by ELIZA then becomes a cornerstone of AIML, which is used pattern-matching rules to recognize the category of user input and then give predefined responses (Wallace, 2003).

Although using AIML to perform dialogue and QA tasks is lightweight and easy to configure (Satu et al., 2015), it has drawbacks. According to Thorat and Jadhav (2020), When the input pattern does not meet any predetermined rules, the rule-based orientation chatbot is inefficient in answering the questions. The study stated that writing rules for varied situations are a time-consuming task, not to mention the difficulty of covering all possible scenarios. These rule-based chatbots can handle straightforward questions, but they lack the ability to handle complicated ones.

The majority of question-generating systems use a rule-based approach to solve the problem. The steps are as follows: first, obtain the grammatical structure of the given sentence, then conduct sentence-to-question transformation using external templates or internal rules. However, depending just on syntactic representations will only be able to generate questions from low-level abstractions (Keklik et al., 2019).

<u>Related Educational Virtual Assistants – “Avatar-based virtual support system”</u>

In 2015, Harvey and his team developed an avatar-based virtual support system. The system's provided answers to employment-related questions by text and spoken responses. The team also customized the avatar's appearance and voice (See figure 2.1 below).

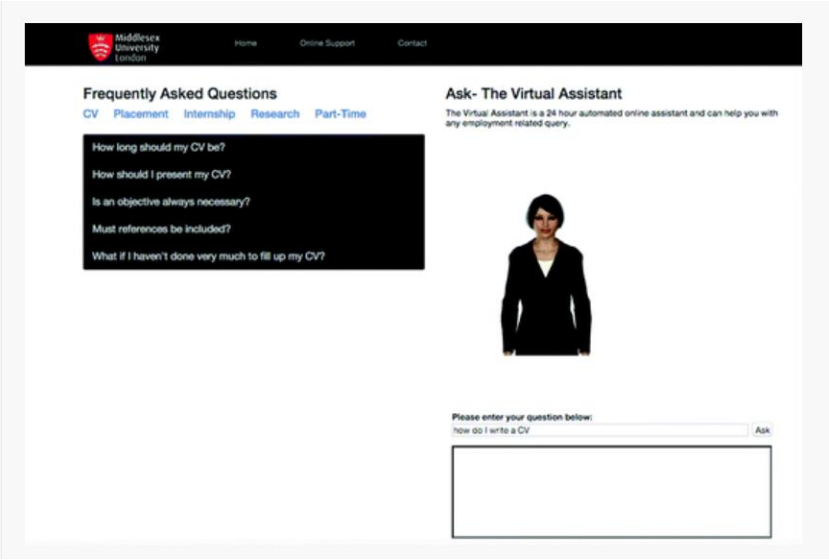


Figure 2.1 shows interface of avatar-based virtual support system (Harvey et al., 2015)

The pilot system got mostly positive feedbacks, as the avatar responded relevant and useful answers to users' inquiries. The verbal response of the avatar gave users the impression that someone was physically presented to assist them.

The system employed Artificial Intelligence Markup Language (AIML) which is a rule-based approach to achieve question-answer interaction. According to Trivedi (2019), while AIML makes the process of conversation modeling reasonably simple, it has an unavoidable flaw in that it is incapable of producing responses on its own. AIML generates a prepared answer or input word substitute by matching a series of words (Trivedi et al.,2019). Programmers must create rules for each query category and offer a knowledge base for replies. (Harvey et al., 2015). If the developed system needs to cover more questions, using AIML creates excessive repetitive works for developers.

2.2.2 Chatbot-building Platforms

Another common approach to performing dialogue and QA is using chatbot-building platforms. Generally, those platforms use "Contextual Reactive Pattern", such as Google's Dialogflow, IBM Watson Assistant, and Amazon Lex (Srivastava et al.,2020 July; Srivastava et al.,2020 September). The "Contextual Reactive Pattern," according to Srivastava and Prabhakar (2020, September), defines a context by declaring a set of Intents and Entities and providing some customer utterances

as samples. It is different from the rule-based approaches because the intent classification requires training phrases while rule-based approaches match a defined pattern.

Chatbot-building platforms have a variety of advantages, including the need for a small number of training data and pre-trained common types of conversation such as greeting and error handling (Srivastava et al.,2020 July). Nonetheless, they share the same flaw as the rule-based method. Because users need to indicate intent and provide related training phrases, a complex query associated with several Intents cannot be handled, necessitating the tools to incorporate more sophisticated NLP mechanisms (Pérez-Soler, 2020; Srivastava et al., 2020 July).

Existing Education Related Virtual Assistants – "Jill Watson"

Jill Watson, a virtual teaching assistant, is developed to answer frequently asked questions from students in Georgia Tech class (Goel et al., 2016). She accomplished a question-answering NLP challenge by retrieving an associated answer after categorizing the query. She will answer students' questions if she identifies the answer to that posted question and not yet answered by other human TA (See Figure 2.2 below).

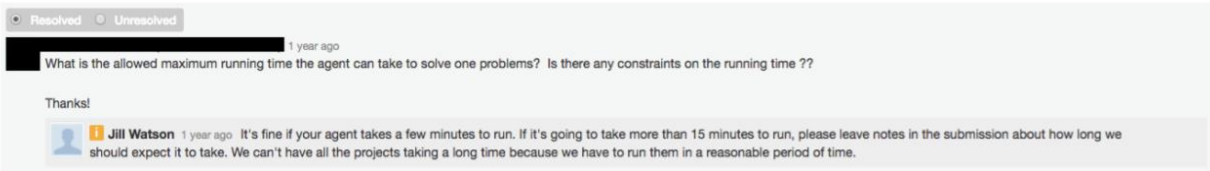


Figure 2.2 shows Jill Watson answered a question on the online forum (Goel et al., 2016)

To create the initial version of Jill Watsons (JW1), Goel and his team (2016) used IBM's Watson, a chatbot builder approach. Watson is a question-answering computer system and provided a platform for user to create conversation service (Godse et al.,2018; Lally et al., 2011). Nevertheless, when the team examined the performance of JW1, they found it had room for improvement as the range of questions JW1 covered was not big enough. The team decided to use their proprietary software and open-source libraries available in the public domain to develop the next generation of JW1.

The performance of Jill Watson shows that an NLP-based virtual teaching assistant is a great automated approach for reducing teaching staff workload and resolving human resource shortages. Despite the success of the initial version of Jill Watson, it is designed to answer questions of one single course. If the course changes the syllabus massively, Jill Watson may not be applicable in the teaching team.

2.2.3 Deep Learning

Deep learning outperformed traditional machine learning approaches such as SVM and logistic regression (Lauriola et al., 2021; Young et al., 2018). Deep learning can learn automatic feature representation in multi-level, whereas typical machine learning-based NLP systems rely primarily on hand-crafted features, which take time and are frequently incomplete (Young et al.,2021). Major deep learning-related models, including Recurrent Neural Networks (RNN), Long short-term memory (LSTM), and Transformer, will be examined.

The RNN is a sequential model dealing with sequential data (Du et al., 2013). RNN works on the principle of saving a layer's output and feeding it to new input to predict the following output, according to Hussain et al. (2019). However, if the sequence is long, the accumulation of information may cause the gradient to disappear or explode (Manaswi,2018; Young et al., 2018). As a result, simple RNN is constrained by slow training time and long-range sequence dependences.

To solve the shortcomings of RNN, LSTM is introduced as a variation of RNN. LSTM tackles the vanishing or exploding gradient problem by avoiding the long-term dependency problem. It is attributed to the added memory cells and gates, allowing cells to recall past information for long periods. The flow of information is controlled using input gates, forget gates, and output gates (Fu et al., 2019; Hussain et al., 2019; Manaswi, 2018; Sherstinsky, 2020). Despite its achievements, the LSTM was slower than the RNN and didn't perform parallel computations (Acheampong et al., 2021).

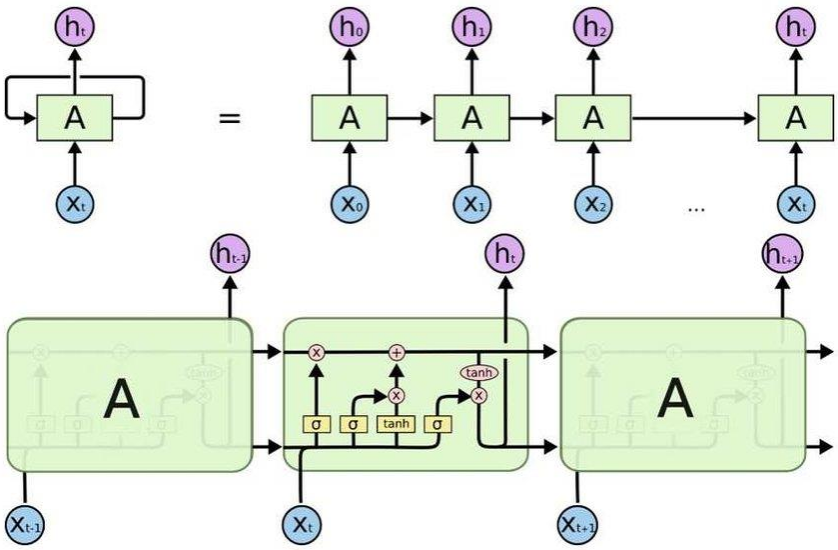


Figure 2.3 RNN and LSTM comparison chart (Fu et al., 2019)

The Transformer was proposed to alleviate the shortcoming of RNN and LSTM. The Transformer introduced by Vaswani and his team members (2017) is the first model to use only self-attention to handle its input and output rather than sequence aligned RNNs or convolution. To convert the decoder output to anticipated next-token probabilities, the Transformer uses an encoder-decoder structure with a linear transformation and softmax function. Before inputting embedding sequence to the encoder or decoder, the position information of a word in a sequence will taken into account. The encoder is composed of 6 identical layers and each layer has multi-head attention and a feed forward network. The decoder shared similar structures to encoder, but have one more masked multi-head attention sub layer. The self-attention layers have less total computational complexity and allow parallelization. Also, it resolve the long-range dependencies issue in the network. Based on the architecture of transformer, different kinds of pre-trained models were developed, such as BERT, RoBERTa, GPT and T5 etc (Devlin et al., 2018; Liu et al., 2019; Radford et al.,2018; Raffel et al., 2019).

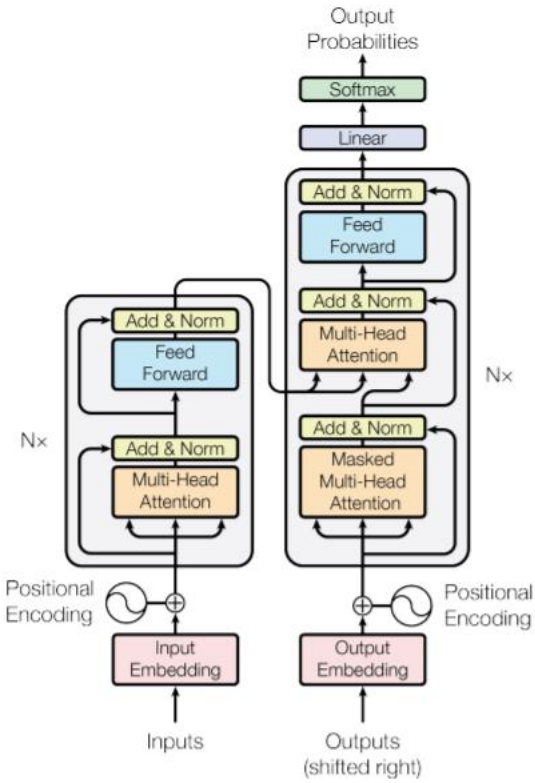


Figure 2.4 Transformer model architecture (Vaswani et al., 2017)

<u>Existing Education Related Virtual Assistants – "Smart campus virtual assistant"</u>

Chiu and his team (2020) developed a smart campus virtual assistant to answer students' common questions. The system used the deep learning approach by developing CNN and LSTM. With emerging speech recognition technology, Chiu's virtual assistant takes users' recordings as input and returns responses by voice and text. For the responses, the emotions of the user would take into account to provide basic contextual dialogue. This feature makes the virtual assistant more human-like and always offers a friendly and caring response.

Our research is different from the others by developing a less administrative virtual course assistant. Firstly, we followed by deep learning approach by adopting BERT as our primary model. We perform two levels of QA tasks, QG tasks, and SA. Secondly, our course virtual assistant integrates with a web-based course information system that allows instructors to add and update course information. We believe the virtual assistants will bring about a more individualized learning experience for each student, increasing their performance, motivation, and self-regulatory skills.

3. Proposed Design

3.1. System Overview

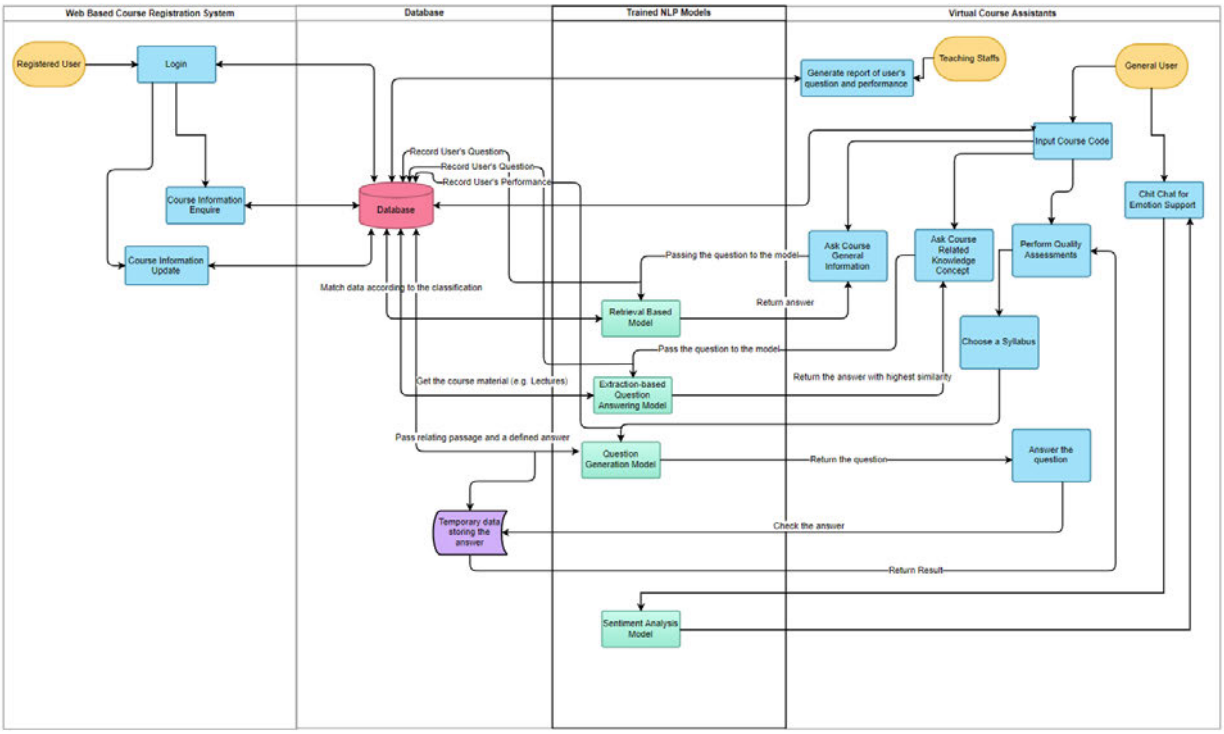


Figure 3.1 The system structure of the proposed solution

The problem definition specifies that the virtual course assistant has two key obstacles: ever-changing data and performing NLP tasks. The first difficulty will be addressed by creating a web-based course registration system, while the second will be handled by using pre-trained language models.

The project uses two platforms and include functions listed below.

Web-based Course Information System	**Virtual Course Assistant**
1. User Login 2. Course Enquiry 3. Course Registration 4. Course Update 5. Generate Performance Report	1. Course General Information Enquiry 2. Course Related Knowledge Concepts Enquiry 3. Quality Assessment of Multi-courses 4. Sentiment Analysis on Student Feedback

As course information may frequently change in response to teacher requests or other academic purposes. The database would need to be updated regularly. As a result, we will roll out a platform for responsible users to add, modify, and update course information.

The NLP tasks will be tackled by constructing three kinds of models, including multiclass classification, extractive based question answering and question generation. The multiclass classification model will be applied to the functions: Course General Information Enquiry and Sentiment Analysis on Student Feedback. Then, the extractive based question answering will applied to Course Related Knowledge Concepts Enquiry function. Finally, the question generation model will applied to the Quality Assessment of Multi-courses functions.

A Use-case diagram that describes the high-level functions and scope is shown below (See Figure 3.2). The diagram also identify the interactions between the system and its actors.

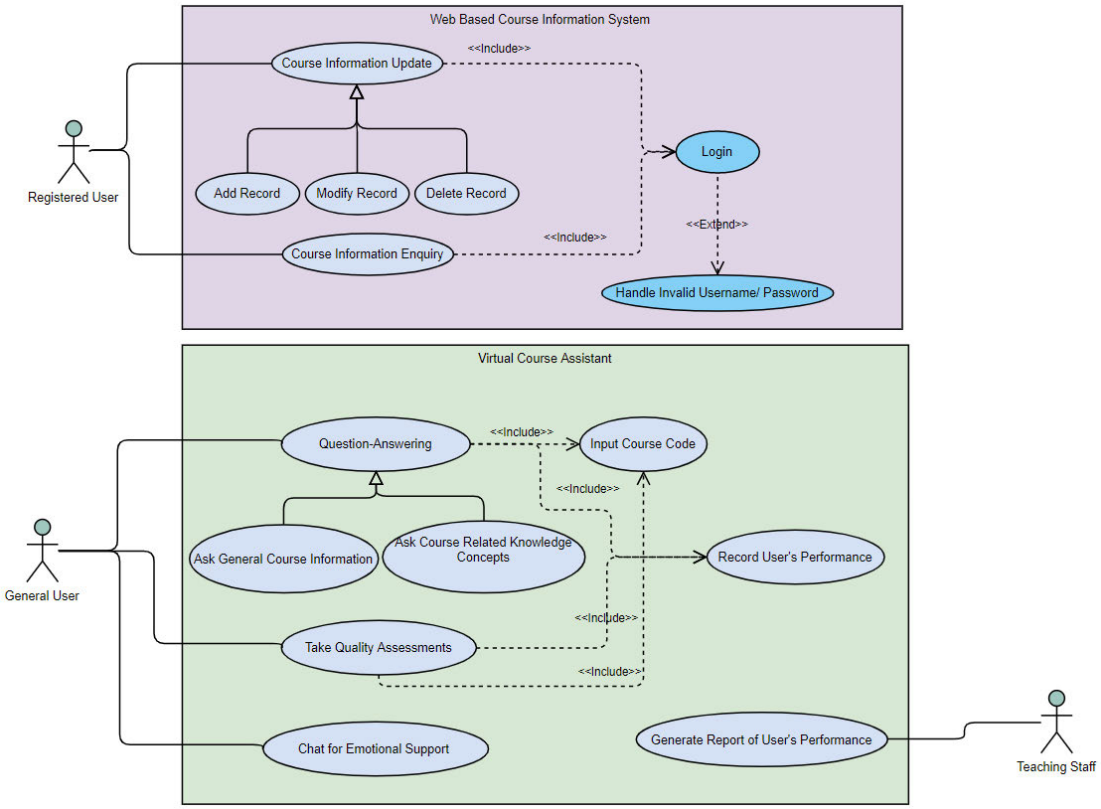


Figure 3.2 Use Case Diagram

3.2. Data Collection Method

Three NLP activities are performed by the NLP virtual course assistant, including two levels of QA, and SA. These are closed domains for QA. Course information, such as lecture and tutorial notes, is required in conjunction with the course materials. We will collect data for the database because there is no relevant dataset online that meets our needs. To train and validate the NLP models, we will also develop our own training and testing data.

The general course information data is collected from the City University of Hong Kong (CityU) below.

- http://www.cityu.edu.hk/catalogue/ug/current/catalogue/catalogue_UC.htm?page=B/B_course_index.htm
- https://banweb.cityu.edu.hk/pls/PROD/hwscrssh_cityu.P_GetCrse .

The course materials are collected from my past enrolled course in CityU.

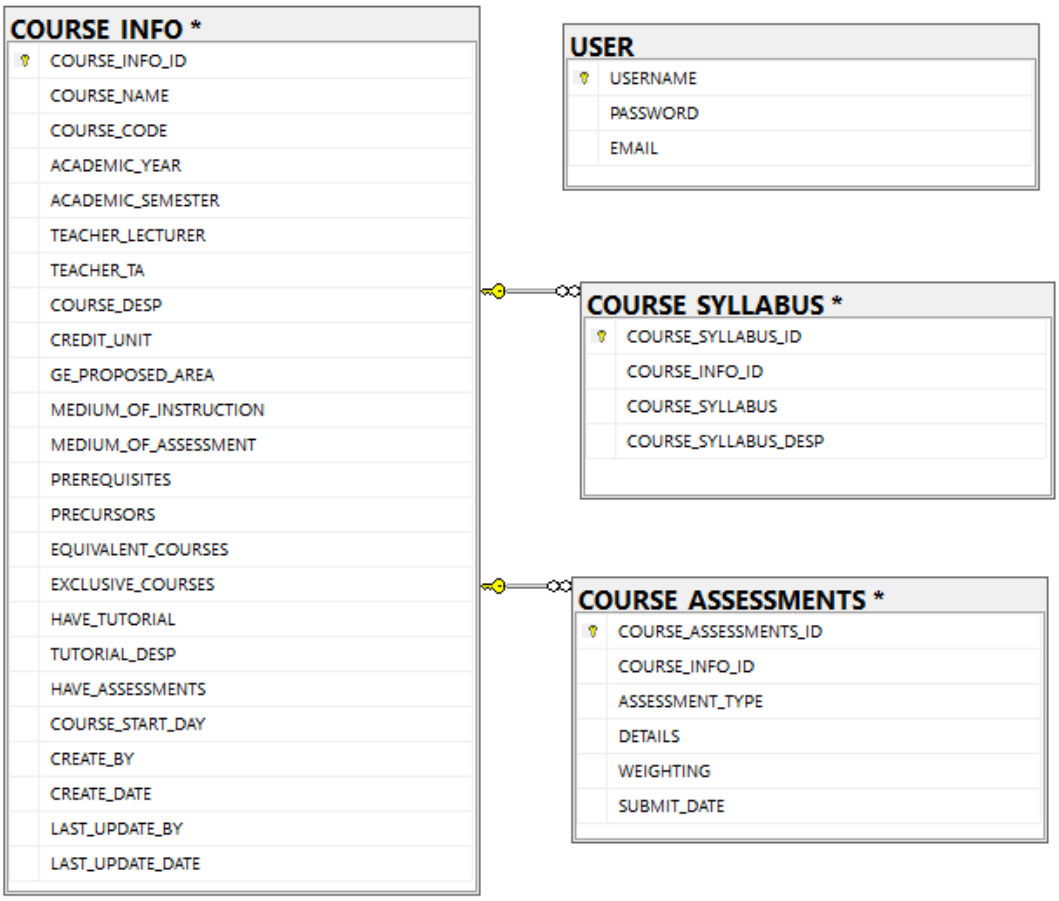


Figure 3.3 database diagram of the project

On the other hand, the SA task is the open domain. We will adopt the Twitter dataset (https://www.kaggle.com/c/twitter-sentiment-analysis2/data) to train the NLP model.

3.3 Performance Evaluation Design

Unit Testing

During the system development, we will perform unit tests for each developed program according to the function specification defined in section 3.1.

System Test

Upon completion of program development, we will carry out a System Test to ensure that each individual program is functional. Also, we evaluate their performance with the below measurements.

Web Based Course Registration System

Ghattas et al. (2020) summarized 11 website performance measurement metrics and created an equation to evaluate website performance. We will evaluate the website according to the listed metrics.

Table 4.3: Website Performance Measurement Metric

Web Metric
Response Time
Load Time
Broken Links
No. of Request
page Size
mark-up validation
design optimization
Page Speed
Start time render
Connection time
DNS lookup

Figure 3.4 Website Performance Measurement Metric summarized by Ghattas et al in 2020

Vatansever et al. (2018) listed the online web-diagnostic tools for data collection about the website performance. We will reference the list and perform data collection through these tools.

Online Web- Diagnostic Tools for Data Collection

Quality Criteria	Online Web Diagnostic Tools
Traffic	www.alexa.com
Page Rank	https://www.pagerank.net/pagerank-checker/
Design Optimization	tools.pingdom.com
Load Time	tools.pingdom.com
Response Time	www.websiteoptimization.com
Mark Up	http://validator.w3.org/
Broken Links	Xenu's Link Sleuth

Figure 3.5 List of online web-diagnostic tools for data collection (Vatansever et al., 2018)

Virtual course assistant

The virtual course assistant will perform two levels of QA tasks and QG tasks. As all of them are closed domains, We will construct our testing data to test the model and obtain the classification accuracy score. Also, we will evaluate the model's performance using the Twitter dataset for the SA task.

System Integration Test

When the System Test passes, we will perform System Integration Test to have overall testing of the complete system. We will test on different scenarios and validate the changes made on the Web-Based Course Registration System will not lead to errors in the other components, virtual course assistant.

3.4. Software Development Approach and Stages

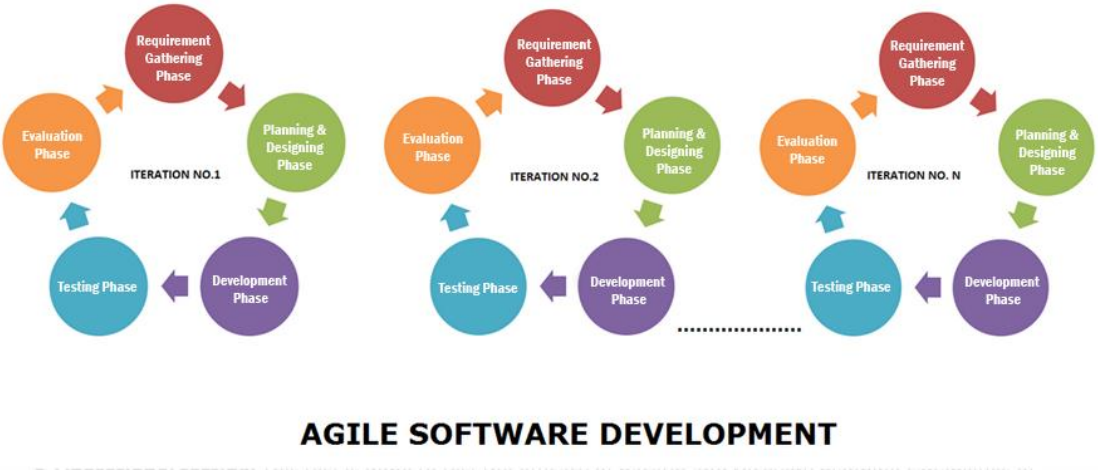


Figure 3.7 Diagram of Agile Software Development

Agile software development is adopted for this project. Programming tasks are distributed to four iterations in execution phase. The work plan is composed of 6 stages:

<u>Phase 1 : Plan</u>

Stage 1 – Project Planning

<u>Phase 2 : Execute</u>

Stage 2 – Database Construction and Web-based Course information System

Stage 3 – Part of Virtual Course Assistant Chatbot functions, Improvements to Stage 2 deliverables and Interim report I

Stage 4 – Part of Virtual Course Assistant Chatbot functions, Improvements to Stage 3 deliverables and Interim Report 2

Stage 5 – Part of Virtual Course Assistant Chatbot functions, Improvements to Stage 4 deliverables and Final Report

<u>Phase 3 : Close</u>

Stage 6 – Presentation and Product Demonstration

4. Detailed Methodology and Implementation

4.1. Encoder Model

Bidirectional Encoder Representations from Transformers (BERT)

BERT by Devlin et al. (2019) is one of the significant models which evolved from the transformer architecture. This model is designed to build deep preliminary learning of bidirectional text representation for machine learning models. BERT's model architecture is almost identical to the original implementation of transformer (Vaswani et al.,2017). The benefit of this model is its ease of use and faster training time. Since it only requires adding one output layer to an existing neural architecture to generate text models that outperform the inaccuracy of all existing ones in several natural text processing problems (Koroteev, 2021). It used encoders in a transformer as a substructure to pre-train models for NLP tasks like SA, QA, and text summarization (TS). BERT executes these tasks in two stages: pre-training for language comprehension and fine-tuning for a particular activity (Acheampong et al., 2021). Further information about BERT fine tuning tasks that described in Vaswani et al. (2017) will be explained in the paragraphs below.

A number of downstream tasks could be handled by the BERT. The Multiclass classification and extractive based question answering job will be used in this project. The BERT input representation could express both a single sentence and a pair of sentences in a single token sequence. For each downstream task, one could simply add task-specific inputs and outputs into BERT and fine-tune all of the parameters end-to-end.

4.1.1 Multiclass Classification Model

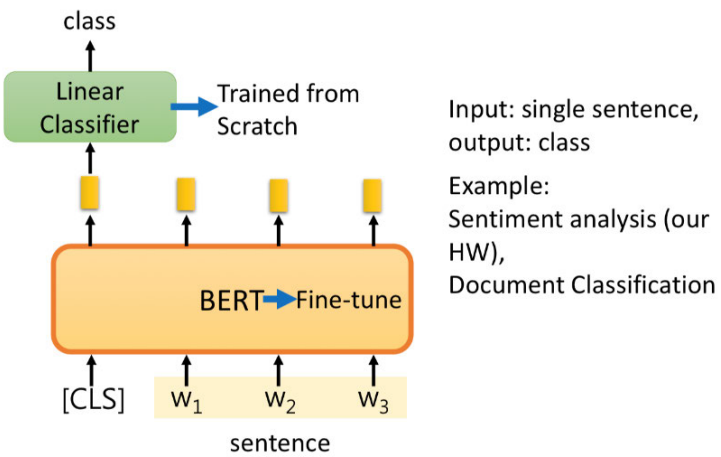


Figure 4.1 shows the architecture of the Retrieval Based Model by BERT (Lee, 2021)

The main task of the multiclass classification model is to classify a correct label of a text. A linear classifier that outputs a certain number of labels must be added to complete the task. Then, one

could embed a text to BERT as input, and then the model will output a tensor of the probability of each label. The highest one is the predicted label. Based on the key concepts of this model, a sentiment analysis model needs to replace the linear classifier with a sentiment classifier. Thus, the modified model will be able to classify the sentiment of a text.

4.1.2. Extraction-based Question Answering Model

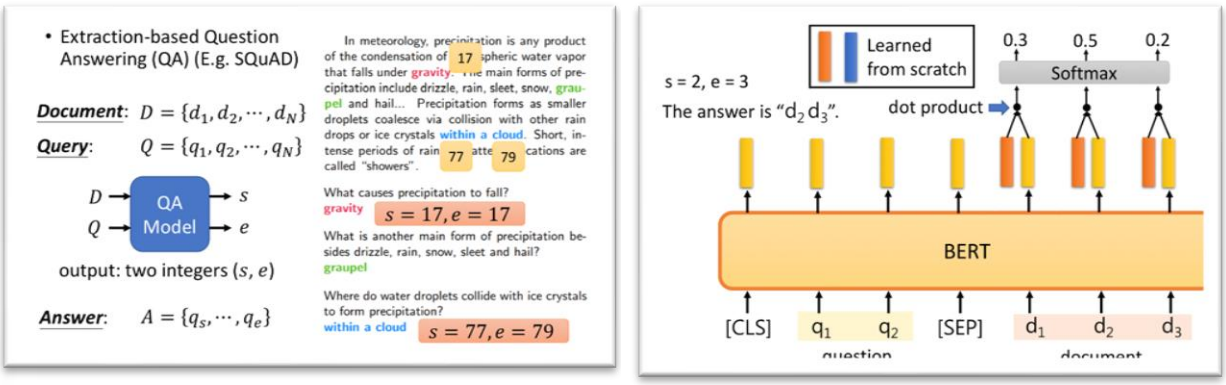


Figure 4.2 shows the architecture of the Extraction Based Question Answering Model by BERT (Lee, 2021)

In the extraction-based question answering task, given a question and a passage, the model needs to identify the answer from the passage. This task does not generate an answer by model. Instead, it identifies which word or sequence is the answer from the given passage. This task requires passage and question pack together into a single sequence as the input of BERT. Firstly, we add [CLS] token at the start of the sequence, separate passage and question with a special token [SEP]. Secondly, a learned embedding needs to add to every token indicating whether it belongs to the passage or the question. Then, the model will output a tensor of numbers, which indicates the start digit and end digit of the answer in the given passage. If the start digit is larger than the end digit, the model could not find the answer from the given passage. This model will apply to the "Answer Course Related Knowledge" functions and generate the most suitable response.

4.2. Web-based Course Information System: Interface Design

4.2.1. Overall Structure

The web-based course information system is a web application designed with a broad shallow tree structure, and all the pages can be accessed on the course enquiry page. The detailed design of these functions will be discussed in section 4.1.2.

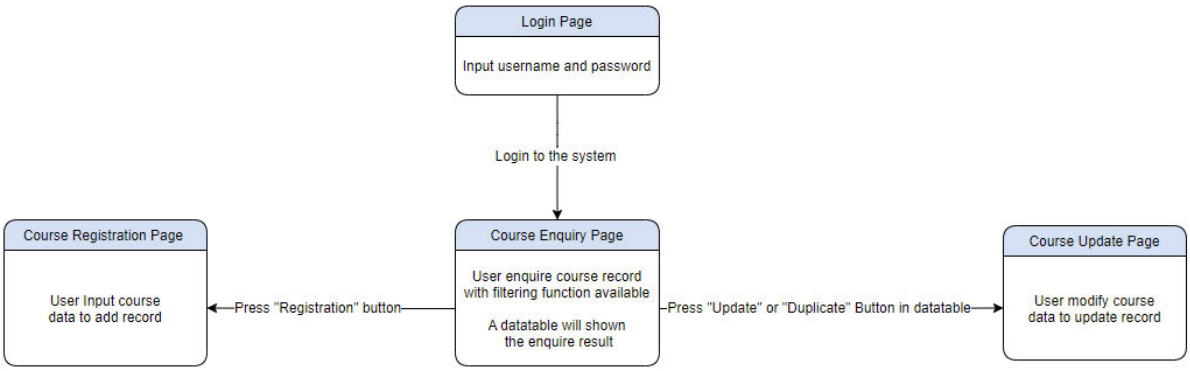


Figure 4.3 Structure and Relations of the Web Interface

The application has a maximum depth of 1 and a breadth of 3 structure. Typically, a user can access various pages with a single click. It reduces the amount of time a new user must spend learning how to utilize the web application. The application enables users to have a higher speed, accuracy, and preference in operations with this broad shallow tree structure.

4.2.2. User Experience Analysis

Firstly, user will need to fill in the forms in this web system. As he might not able to memorize all the details, so he may need to open another website or document for reference. Under this scenario, our web system should be responsive. Therefore, when user resize the window of browser, he can still fills in the form easily, without any context hidden.

Secondly, user may search more than 1 record in the enquiry page. For user convenience ,a clear button that can reset all the filtering input to its default state should be created. It can speed up the time of searching of frequent users.

Last but not least, the web system consists of many texts for user to read, it is necessary to enhance the readability. For example, the font colour should be a contrast colour of the background. Also, the functional components should have some response when user move the cursor on them, it helps to distinguish the functional components and non-functional components.

4.3. Web-based Course Information System: Functionality

The web-based course registration system is developed to handle data. Therefore, responsible users could modify information when needed. In the long run, this less administrative system

could eliminate the admin or developer works. We will describe details of the functions included in the system.

User Login

Account will be assigned to related users to log in to the system. The measure is necessary for the following reasons. Firstly, they are a security measure designed to prevent unauthorized modification and access to the data. Secondly, they are used to trace the responsible user of managed records. Also, while the user login, the inputted password will be hashed. Only the password's hash value will be retained in the database for security purposes.

Course Enquiry

After the user logs in to the system, an enquiry page will be displayed in default. Users can search the record of a specific course and view the details. In the searching criteria area, user could tick the checkbox of “Record(s) Created By You” and perform searching, the result will only display the records that created by user. In the datatable, each column header has a sorting icon, if user clicks it, the whole datatable will sort by that column in alphabetical order. The enquiry page provides users an interface to view all the records saved in the database.

Course Registration, and Course Update

To register a new course record, users can click on the "Register New Course Record" button to access the Course Registration page. The page will show a form for users to fill in the course information, such as Course Name, Course Code and Credit Unit. Users could update the course information by clicking on the "View/Update" button in the Action column of the course record. Only the record owner could view the "View/Update" button and access the Course Update page with data filled. The Course Registration page and Course Update page interface are the same to maintain consistency and lower the rate of error by users.

Since we allow users to modify data independently, it is necessary to maintain data integrity by only allowing the record creator to modify the record. However, consider a scenario, in a different academic year, the teacher in charge of the same course may change. The new teacher will need to create a new record to update the course information. While most information might remain the same, we will enable a duplicate function. Then, a course registration page with filled-in data will be displayed. Thus, users can modify and create a new record quickly.

Generate Performance Report

Web application users, primarily teachers or teaching assistants, could sign in to the web application and search the course data record. In the action column of the owned course record, a generate report button will be displayed. The report will use openpyxl to write the data to an excel file. Only the course record owner could generate the report about the performance of the virtual assistant user, mostly students. The restriction is to keep data confidential to the authorized user.

The generated report will contain the below information:

1. The general course information questions received by the virtual course assistant. The report will calculate the number of questions according to its category and sort the category from most appearances to fewer appearances.
2. The course-related knowledge questions received by the virtual course assistant. The report will calculate which lecture chapter is related the most.
3. The feedback of students to the course. The report will conduct sentiment analysis to each feedback and calculate the total number of each sentiment.

The functions of virtual assistant require different datasets to train the corresponding model, yet the defined scope is CityU's course. Two of the datasets need to be constructed by ourselves.

Required Techniques

HTML, jQuery, JavaScript, data tables, CSS, and Bootstrap are used to develop front-end functions and style the website. When designing the functions in such works, system design concepts will also be taken into account. Python and SQLAlchemy will be used to develop the backend functions. Adding, updating, deleting records are some examples that the backend will deal with.

The database will be designed by constructing database diagrams. Also, rules like normalization will be applied. MS SQL will be used to manage the database. Knowledge of setting up the connection with other software is necessary. Python will be used to scrape data from documents in batch and save them to the database.

4.4. Virtual Course Assistant: Dataset Construction

The functions of virtual assistant require different dataset to train the corresponding model, yet the defined scop is cityU's course, two of the dataset need to construct by ourselves.

CityU General Course Information Dataset

This is a dataset of 258 questions about general course information enquiry, labeled by category. The dataset includes 2 columns, the Question and Category.

Date Category	**Data Type**
Question	A question asking general information of course (String)
Category	The referred information category of the question (String) There are 11 categories: "ASSESSMENT_INFO", "COURSE_DESP", "COURSE_NAME", "COURSE_START_DAY", "CREDIT_UNIT", "COURSE_REGISTRATION", "GE_PROPOSED_AREA", "TUTORIAL_INFO", "MEDIUM", "TEACHER", "COURSE_SYLLABUS"

Tables 4.1 shows the dataset attributes

There are a few steps to create the dataset. Firstly, we need to define the course's general information. Course general information refers to non-knowledge related information but is frequently asked by students, such as assessment weighing and assignments requirements. Research needs to be conducted to classify which kinds of information is general course information. We need to study the Detailed Course Information pdf provided by CityU. After studying some of that information, 23 categories of those general course information were summarized. According to those categories, some potential questions from the students were brainstormed. Each category has 2 - 5 questions. However, after the first model training, the performance is inferior. To solve the problem, some of the labels were grouped, and the data sample of each category was increased. Therefore, the dataset will have 11 categories, and each category will have at least 20 data samples.

<u>Python Course Question Answering Dataset</u>

The data content includes 6 columns, including Context, Question, Answer, Answer Start, Lecture Chapter and Source.

Data Category	**Data Type**
Context	One of the short passages from the lecture notes. The context should be one of the key topics in the lecture note. (String)
Question	A question related to the context (String)
Answer	Answer found in the context (String)

Answer Start	The position of the first character of the answer in the context. Space and punctuation marks count as one character as well. (Integer)
Lecture Chapter	The lecture chapter of the context (Integer)
Source	Source of the context (String)

Tables 4.2 shows the dataset attributes

This dataset is referenced on the SQuAD dataset. To construct this dataset, we need to check if the lecture notes format is suitable or not. Commonly, the lecture notes are in point form, use many images to replace text and use tables instead of passages. Those formats are unsuitable for our data set, so we need to rephrase the lecture notes to a passage. After rephrasing the lecture notes, we can split the context into a smaller passage and then put it into the Context column. Then, according to the selected context, we need to think of a pair of question and answer, while the answer must be a piece of text that appeared in the context. Thus, yes-no questions are unacceptable, and we need to avoid asking questions with Is there/ Is it, but asking questions with who, what, where, when, or why. After the question and answer pairs are completed, we need to record the position of the first character of the answer that appeared in context.

4.5. Virtual Course Assistant: Environment Set up

The virtual course assistant uses telegram as the communication platform. To enable the assistant, we need to find botFather, an official bot creation bot of telegram. Then, we input the" /new bot" command and input the chatbot name and username. Finally, the botFather will generate a token for users to access our telegram bot API.

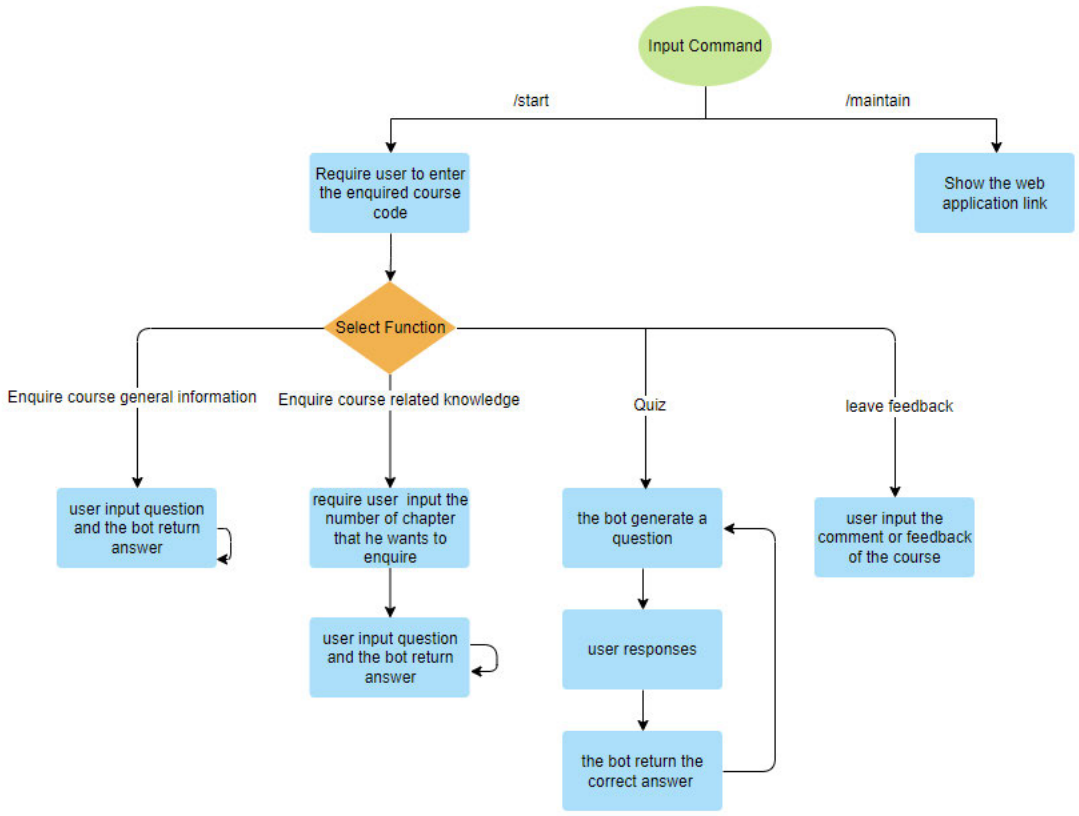


Figure 4.9 shows the conversation flow of virtual course assistant

After getting the token, we need to set up a bot that can constantly receive messages. Firstly, we need to set up commands for the bot through botFather. We enter the command /setcommands, and then we can choose our bot and input the command name and description. There were two commands set up, /start and /maintain. Also, to enable the bot to receive messages and return proper responses constantly, the bot will identify the received message and set the state of indicator to control the response. The /maintain command will return the URL of the web-based course information system, so that users can find the website's link quickly without memorizing the website.

The /start command indicates that the user would like to perform functions about the course. After the user inputs the /start command, the bot will first require the user to input the course code. Prior to user response, the bot will return the input sequence to the backend and check if the course code exists in the database. If not, the bot will return: "there is no such course". If the bot finds the course, the bot will provide multiple choice questions for the user to select the function he needs. The multiple questions are set up with ReplyKeyboardMarkup, so that the options will replace the original keyboard. When the user presses one of them, it enters that function. The backend indicator will be set to the function name, so that the user can constantly ask the same types of

questions. If they want to switch to another course or function, they can input the /start command again. The details of the function will be discussed in the next section.

4.6. Virtual Course Assistant: Functionality

4.6.1. Answer Course General Information

Model Fine-tuning

In this function, a multiclass classification model needs to be constructed to classify the question's category. Firstly, we need to pre proceeds the data. As the data is in string format, we need to encode the category, using a number to replace the original category. Then, a Triage function is defined to tokenize the data at once. A tokenizer corresponding to the loaded pre-training language model is loaded. The triage function will use the tokenizer.encode_plus function to add padding, truncation and add special tokens [CLS] and [SEP] to the text data. After that, a dictionary with the tensor of ids, tensor of mask and tensor of targets will be created. Then, we will define the training parameters, including batch size, shuffle and num_workers to both of the datasets by the Pytorch DataLoader function.

Secondly, we need to self define a model. We created a neural network with the pertained language model followed by a dropout, a dense layer, and a linear layer to obtain the final output. Then, we start training the network by feeding the data to the pre-trained language model. During the training, we calculated the loss by torch.nn.CrossEntropyLoss() and optimise by Adam optimiser.

After the training, we will validate the model with the validation dataset. The procedures are similar to the training function, yet we set the model as eval() model so that the validation steps will not affect the trained network. Finally, we will save the model with the best performance.

We have tried to use five different pre-trained language models with the same classifier and dataset to build the neural network and evaluate the performance of their downstream tasks. We will compare each best model using a different pre-trained language model. The architecture of the models is almost identical. The results will be discussed in section 6.2.

Model Application

The indicator will be set to this name as well, and the bot will send the user the message "what do you want to know?" Users need to input questions that are related to the general course information. Then, we pass this question to the tokenize function, add [CLS] at the start and [SEP] at the end of the question, and then pad 0 to the remaining space. The mask will contain 1 and 0. Then we

pass the ids and mask tensor as parameters to the trained model. The model will output a tensor with the possibility of each encode category. We will select the highest one and map to encode the category to the corresponding label. Then, we will start the query from the database. If the returned label exists in the database, we will select that column of the relevant course, and return the text in the below format:

Category Used in Model	**Template of Response**
COURSE_SYLLABUS	Select COURSE_SYLLABUS and COURSE_SYLLABUS_DESP columns from database and return the results in below template. **Here is some course syllabus information:** **COURSE SYLLABUS:** {first result of COURSE_SYLLABUS} **COURSE SYLLABUS DESCRIPTION:** {first result of COURSE_SYLLABUS_DESP} **COURSE SYLLABUS:** {second result of COURSE_SYLLABUS} **COURSE SYLLABUS DESCRIPTION:** {second result of COURSE_SYLLABUS_DESP} …
COURSE_DESP	Select the COURSE_DESP column from database and directly return the result
COURSE_NAME	Select the COURSE_NAME column from database and directly return the result
COURSE_START_DAY	Select the COURSE_START_DAY column from database and directly return the result
CREDIT_UNIT	Select the CREDIT_UNIT column from database and directly return the result
COURSE_REGISTRATION	Select the PREREQUISITES, PRECURSORS , EQUIVALENT COURSES and EXCLUSIVE COURSES columns from database and return the results in below template. **Here is some course registration information:** **PREREQUISITES:** {result of PREREQUISITES} **PRECURSORS:** {result of PRECURSORS} **EQUIVALENT COURSES:** {result of EQUIVALENT_COURSES} **EXCLUSIVE COURSES:** {result of EXCLUSIVE_COURSES}

TUTORIAL_INFO	Select the HAVE_TUTORIAL and TUTORIAL_DESP columns from database and return the results in below template. **Here is some tutorial information:** **HAVE TUTORIAL:** {result of HAVE_TUTORIAL} **TUTORIAL DESP:** {result of TUTORIAL_DESP}
GE_PROPOSED_AREA	Select the GE_PROPOSED_AREA column from database and directly return the result
MEDIUM	Select the MEDIUM_OF_INSTRUCTION and MEDIUM_OF_ASSESSMENT columns from database and return the results in below template. **Here are some course medium information:** **MEDIUM OF INSTRUCTION:** {result of MEDIUM_OF_INSTRUCTION} **MEDIUM OF ASSESSMENT:** {result of MEDIUM_OF_ASSESSMENT}
TEACHER	Select the TEACHER_LECTURER and TEACHER_TA columns from database and return the results in below template. **Here are some course teacher information:**\ **LECTURER:** {result of TEACHER_LECTURER}\ **TEACHING ASSISTANT:** {result of TEACHER_TA}
ASSESSMENT_INFO	Select ASSEMENT_TYPE, DETAILS, WEIGHTING and SUBMIT_DATE columns from database and return the results in below template. **Here is some course assessment information:** **ASSEMENT TYPE:** {first result of ASSEMENT_TYPE} **DETAILS:** {first result of DETAILS} **WEIGHTING:** {first result of WEIGHTING} **SUBMIT DATE:** {first result of SUBMIT_DATE} …

Tables 4.3 shows the responses returned by assistant

4.6.2. Answer Course Related Knowledge Concepts

Model Application

If user select the function "Ask Course Related Knowledge", the bot indicator will set to this name as well, and the bot will send message "Please enter the number of lecture chapter that you want to enquire." to user. User need to input a number, the bot will check whether the number of lecture exists. If no, the bot will send message "Lecture <input number> not found, please enter another

number.”. If the lecture notes exists, the bot will send message “What do you want to know?” to user. After user inputted a question ***Q***, the bot need to select the possible related passage from lecture notes by the following procedures.

1. By default, a list of common appeared word that clearly is not keyword, such as “is”, “am”, “are”, “who”, “what”, “where” is created. The sequence ***Q*** will split by space , covert each word to lower-case and stored in a list. Then, ***Q*** needs to remove word that appeared in the common appeared word list.

   e.g. ***Q*** = Which operator performs addition in Python? -> ***Q*** = [operator, performs, addition, python]

2. For each remaining word, we need to find all of its appearance in each passage of lecture note. Then, we count the keyword appearance in each passage and store them in a dictionary.

   e.g. passage: “The common operator to form an expression in python are +,-,*,/ . The + operator performs addition. The – operator performs subtraction. The * operator performs Multiplication.”

   dict: {keyword: 9 ; passage: The common operator to form an expression in python are +,-,*,/ . The + operator performs addition. The – operator performs subtraction. The * operator performs Multiplication.}

3. All of the selected passage will pass to a loop function, the function tokenize the passage and user’s original inputted question. Then, the tokenize sequence will pass to the model and the model will return a tensor of start logits and end logits. According to the outputs, a score will be calculated. Finally, the function will compare the score and choose the highest one as result.

4.6.3. Quality Assessment of Multi-courses

When user select the option “Do quizzes”, the bot will set the indicator as “Do quizzes”. Then, the bot will select a set of question and answer related to the course from the database. The bot will send a question and wait for user’s response. When the bot received the response of user, the bot will send the correct answer for user to check. The bot will perform the above procedures until all question are asked or user input command.

4.6.4. Sentiment Analysis on Student Feedback

The sentiment analysis tasks is a multiclass classification which is the same model architecture of “Ask Answer Course General Information”. Thus, the fine-tuning procedures are similar, yet with different dataset and output labels.

<u>Model Application</u>

When user select the function “Leave comment”, the bot will send message “What do you think of this course?”. After getting user response, the bot will send “Thank you” and end the conversation. User’s comment will save to the database and perform sentiment analysis when teacher would like to generate report.

5. Performance Evaluation and Result

As mentioned before, this project uses a Web-based Course Information System and Virtual Assistant to accomplish the goal of develop an all-time applicable virtual course assistant that can provide educational support to teachers and students. This section will present final interface and the evaluation procedures of these two components. On top of that, the data preprocessing, models performance and analysis the results of models will be described detailedly in the subsection of Virtual Assistant.

5.1. Web-based Course Information System Performance Evaluation

5.1.1 Overall Interface

The Web-based Course Information System consists of five pages to serve with the purpose of login, enquiry, registration, update and duplication.

<u>Login Page</u>

When user access the web system, a login page will shown. Only authorized users could login.

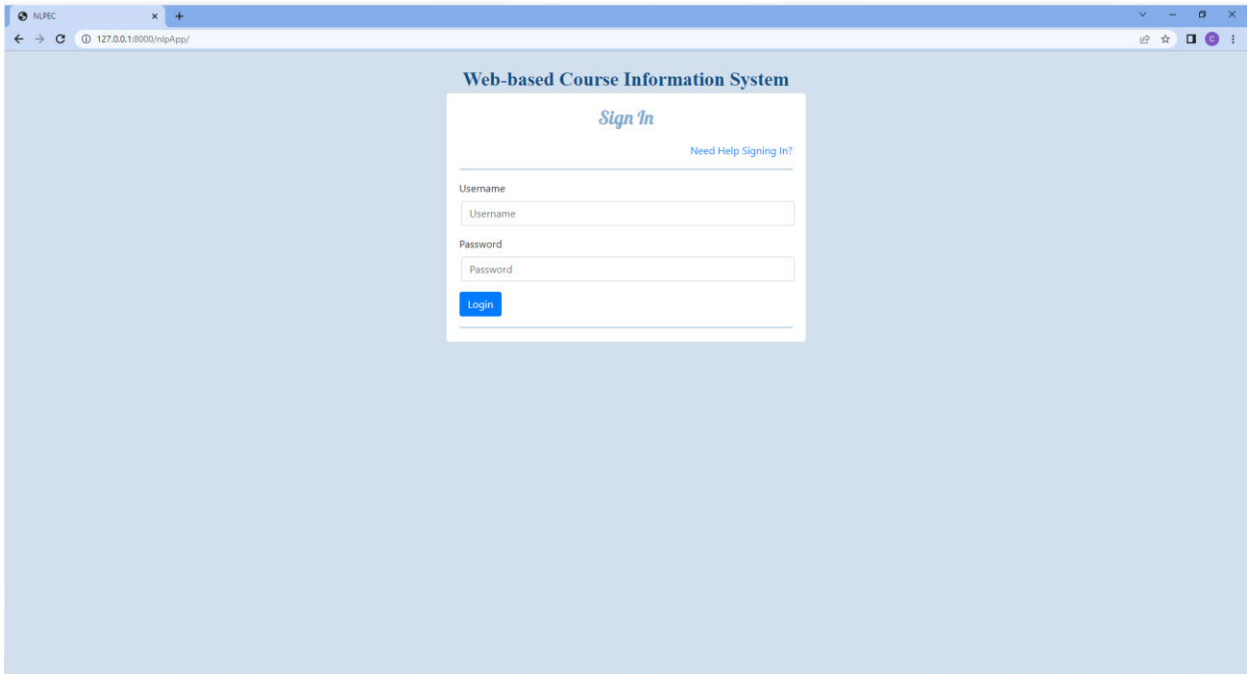


Figure 5.1 shows the login page

Enquiry Page

The enquiry page show after user successfully login. A green bar will display the ID of current user. The web application have user controls ,only the record owner could see the “View / Update” and “Report Generation” buttons , while non record owner will see “View/Duplicate” button if the record is not created by them (see Figure 5.3).

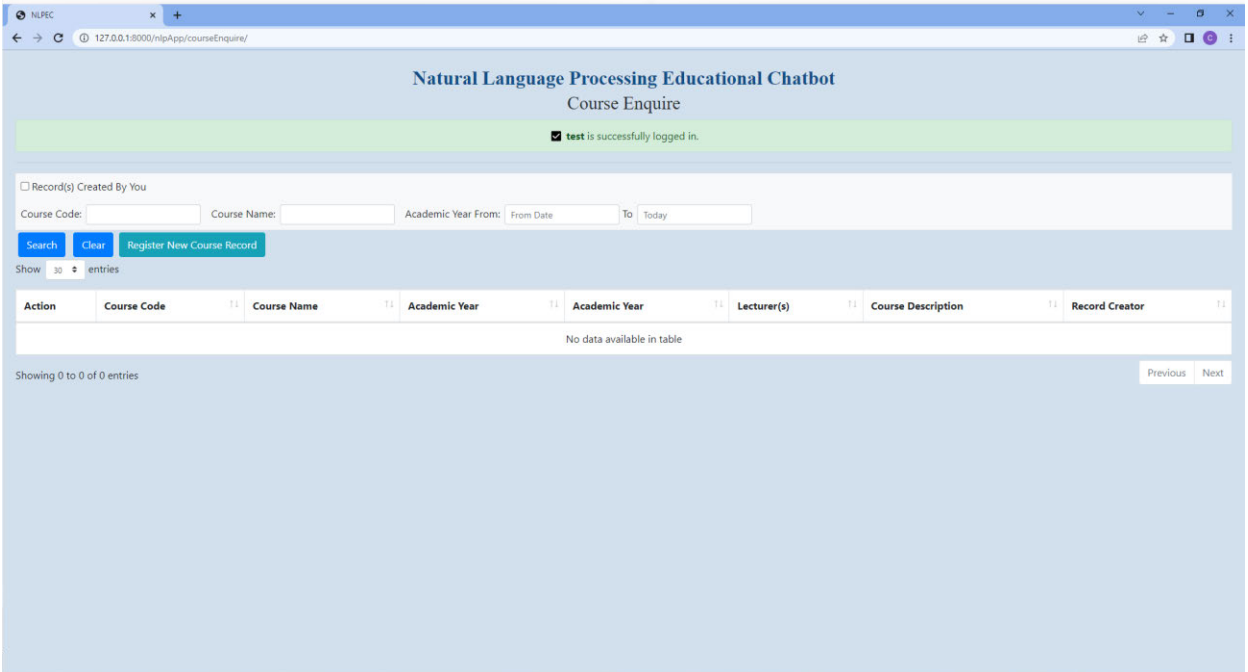


Figure 5.2 shows the default layout of enquiry page

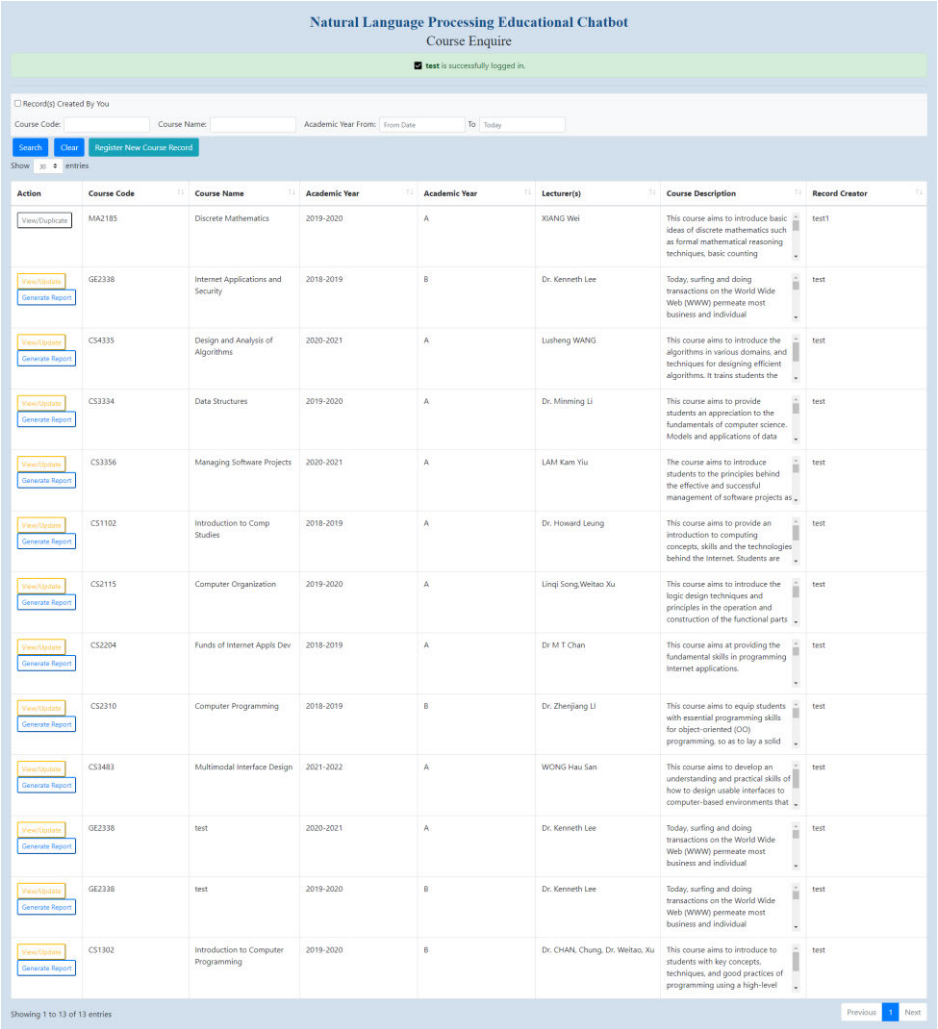


Figure 5.3 shows the default layout of enquiry page

Registration page

If user presses the course registration button in enquiry page, a registration page will open. User can register course in this page.

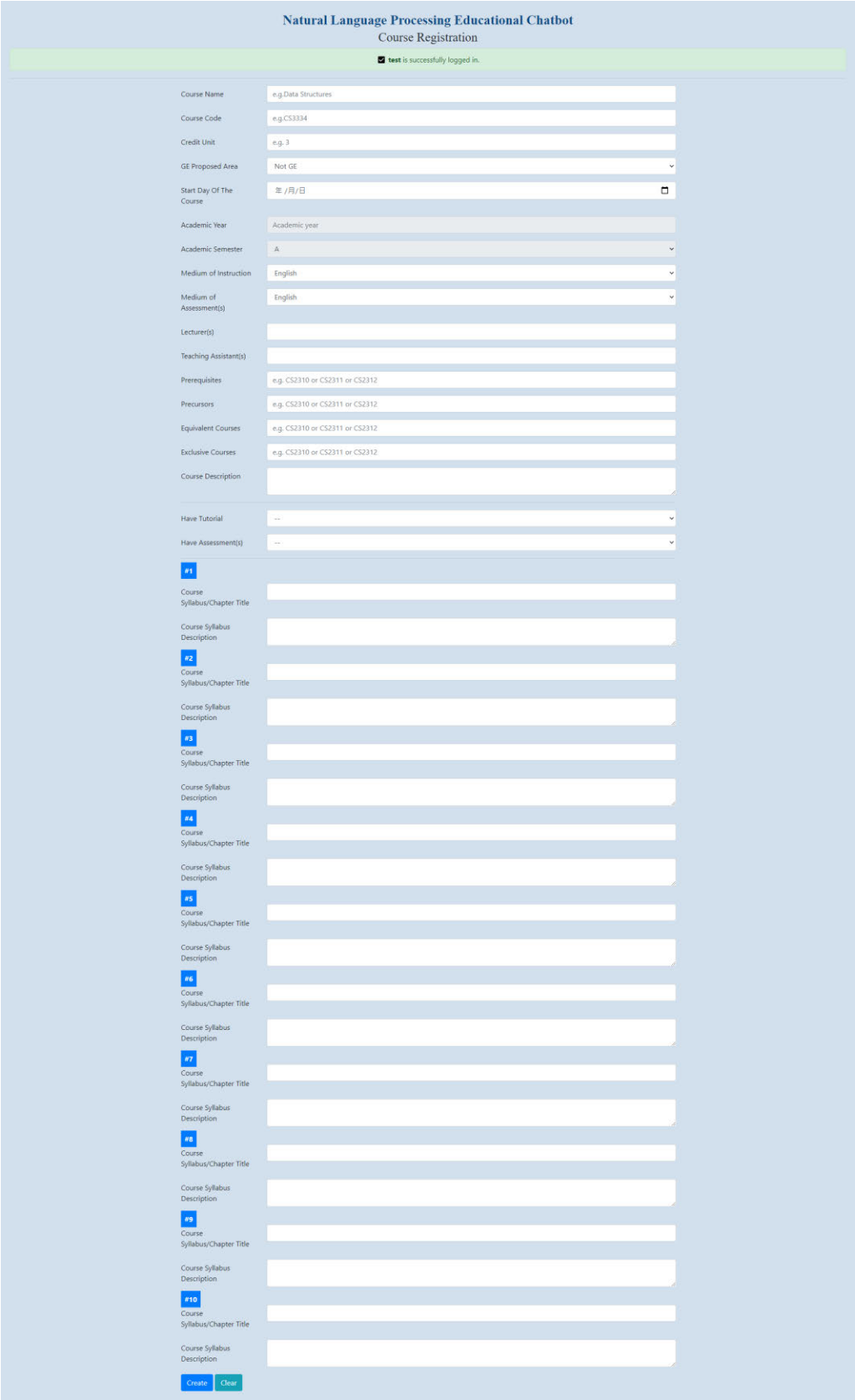


Figure 5.4 shows the default layout of registration page

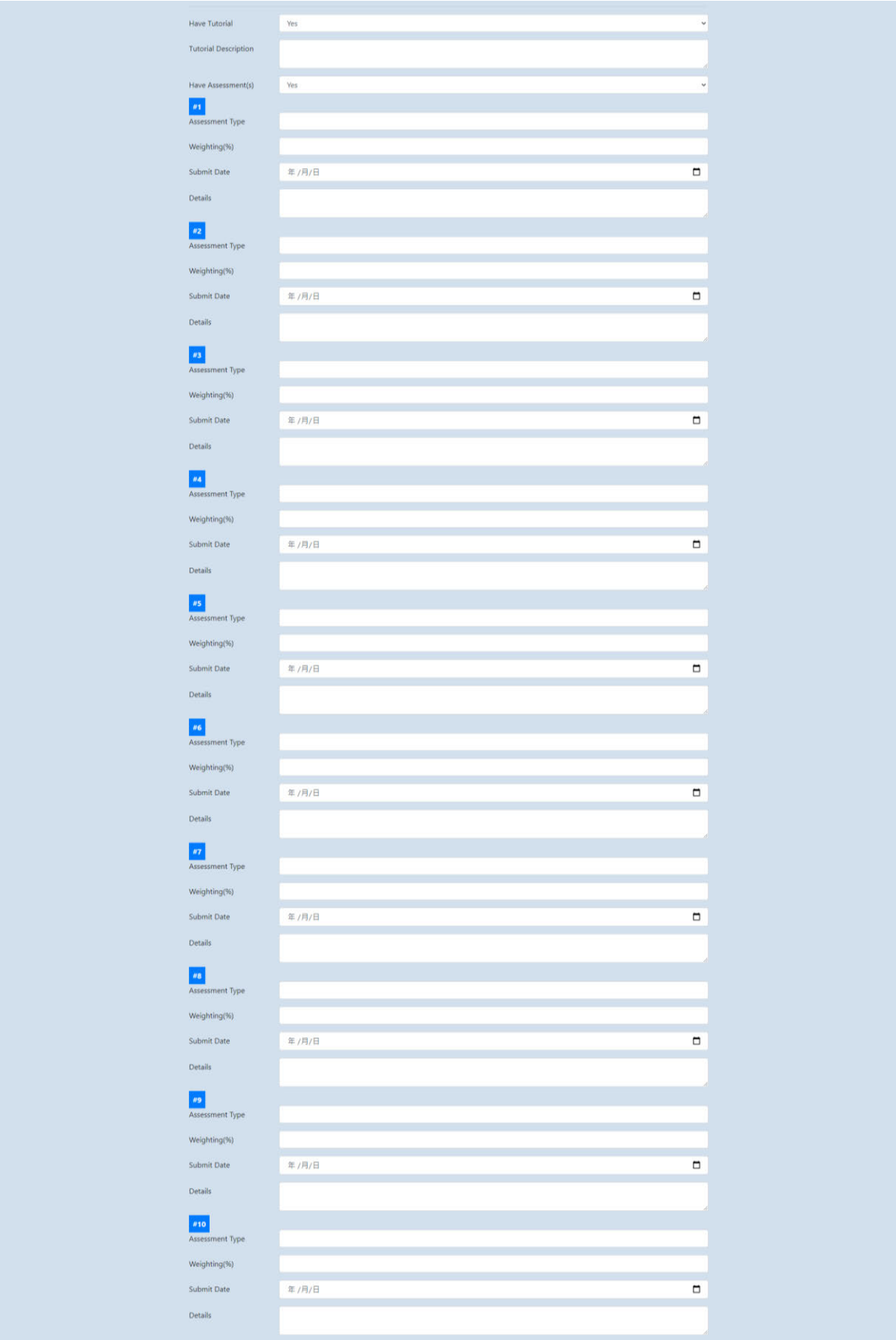


Figure 5.5 shows the layout of registration page after selected value in field Have Tutorial and Have Assessment(s)

Opened Page when clicking the 'View/Duplicate' button

This page have the same layout with registration page.

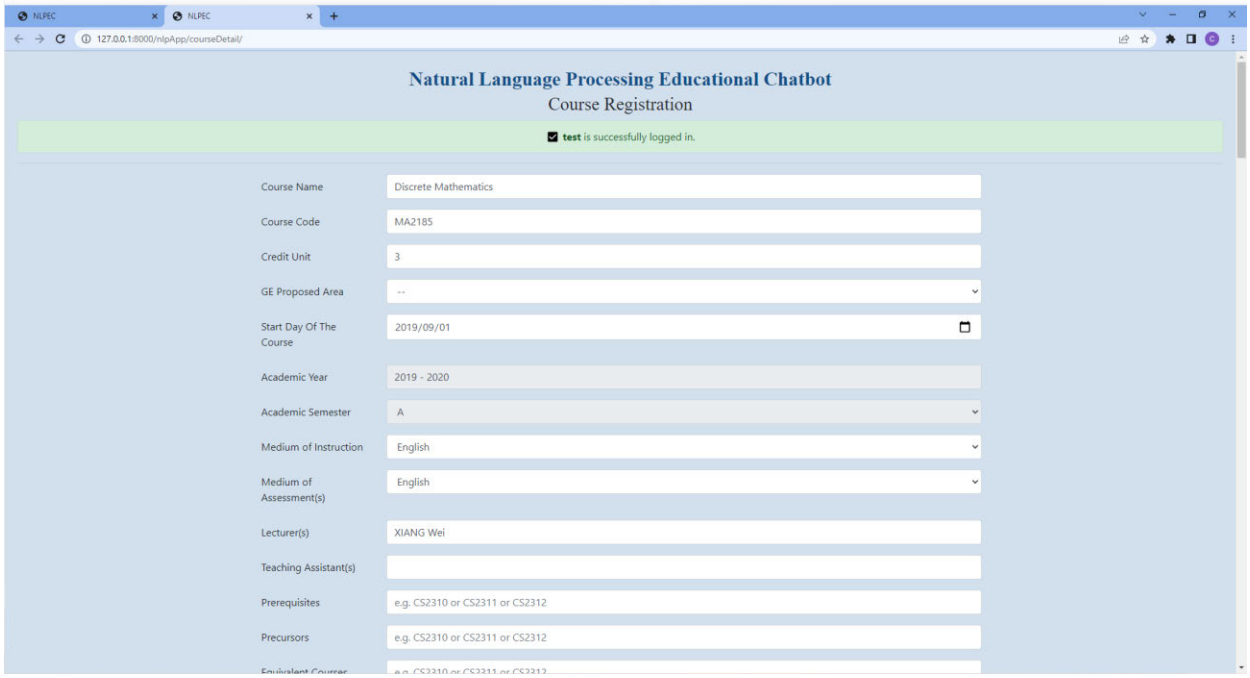


Figure 5.6 shows the layout of Duplicate Page

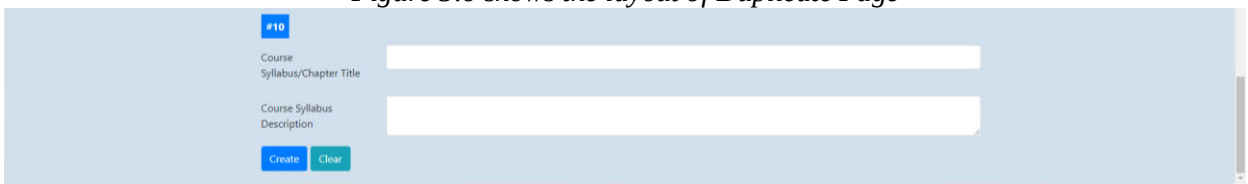


Figure 5.7 shows the Duplicate page only have Create and Clear button in bottom

Opened Page when clicking the 'View/Update' button

This page have the same layout with registration page as well. Yet, it have a 'Update' button in the bottom (see Figure 5.10).

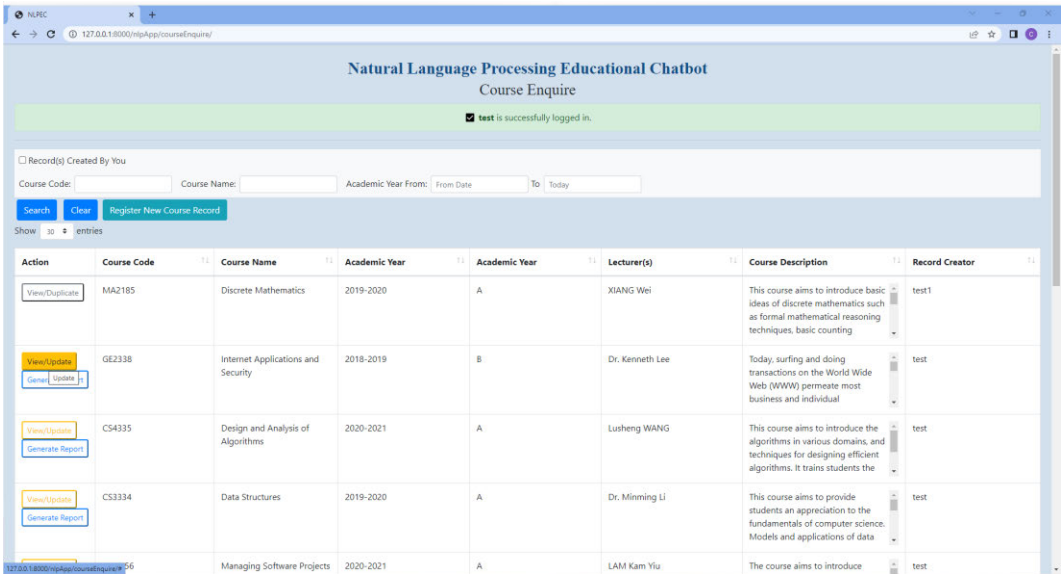


Figure 5.8 shows the way to access Update page

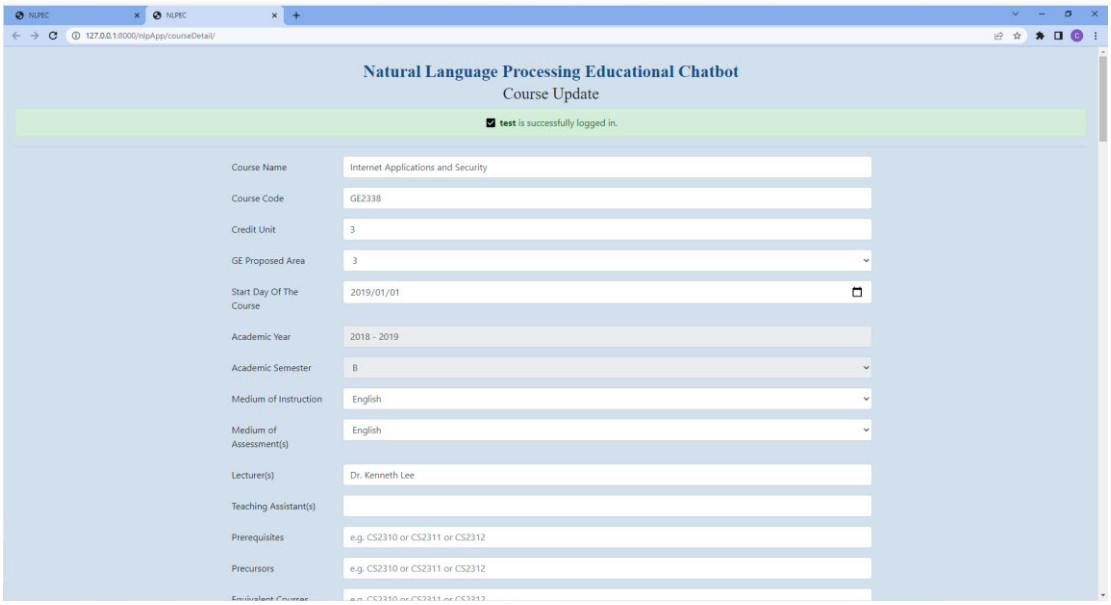


Figure 5.9 shows the layout of Duplicate Page

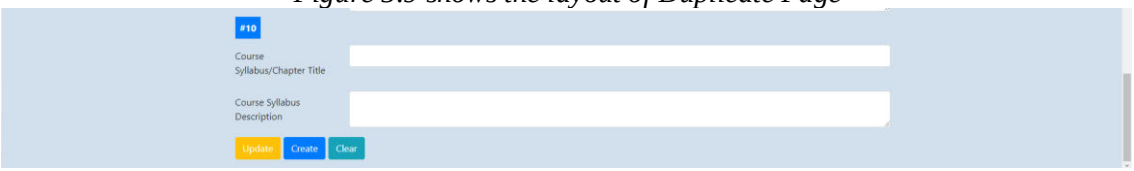


Figure 5.10 shows the Update page have Update, Create and Clear button in bottom

Generate Report

User can generate report of the course statistics through the 'Generate Report' button in enquiry page (see Figure 5.11).

Figure 5.11 shows the way to generate a report

Course Name:	Introduction to Computer Programming			
Semester:	2019-2020 sem B			
Teacher:	Dr. CHAN, Chung, Dr. Weitao, Xu			
Course General Information Enquiry			Total Record:	1
Date	Question Raised	Question Category		
2022-03-27 22:27:26	Do this course have TA?	TEACHER		
Course Related Knowledge Concepts Enquiry			Total Record:	3
Date	Question Raised	Provided Answer	Belonged Lecture Chapter	
2022-03-30 17:17:47	What is the command for addition in intel base cpu	00000011 00000100	1	
2022-03-30 17:18:36	What is abacus	an ancient mechanical computing device	1	
Sentiment Analysis on Student Feedback			Total Records:	3
Date	Student Feedback	Sentiment Analysis		
2022-03-30 17:20:11	Bad	Have room of improvements		
2022-03-30 17:22:00	Good	Excellent		

Figure 5.12 shows the details of report

Originally, the function of generating report is run in the virtual course assistant. Yet, anyone could use the virtual course assistant, and the data of the report would be easily accessed by other. Thus, this function is moved back to the web application , so that only authorized user like teacher could access the report.

5.1.2 Important layout s

Responsive Design

When user resize the window of browser, the position of text and label will swift accordingly.

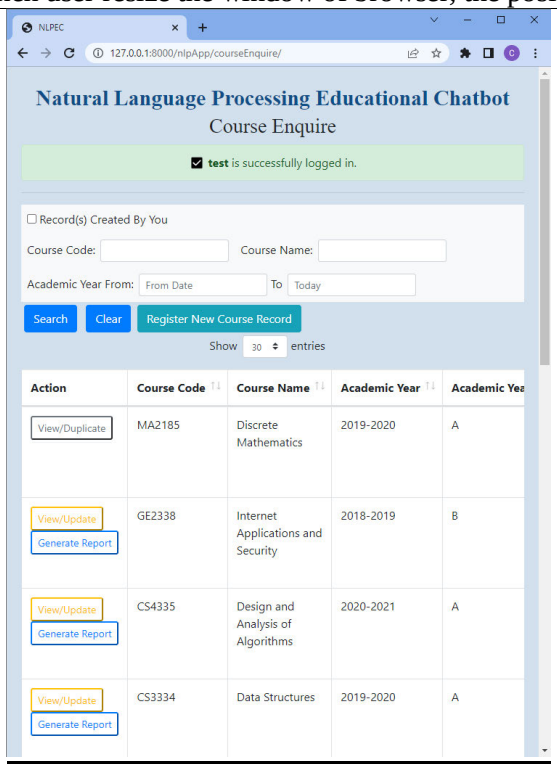


Figure 5.13 shows the responsive layout of enquiry page

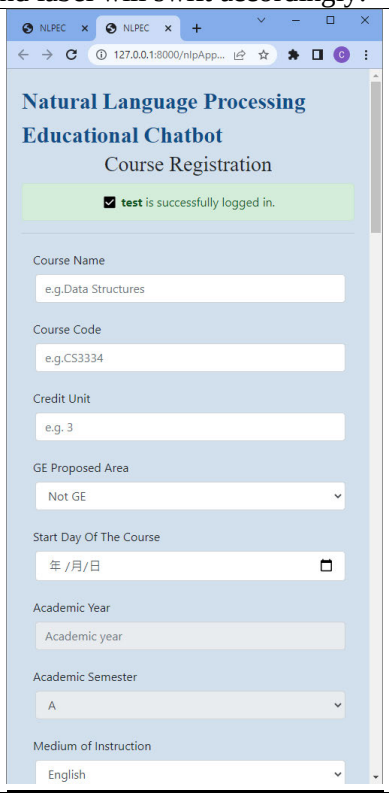


Figure 5.14 shows the responsive layout of registration page

Consistent layout of pages

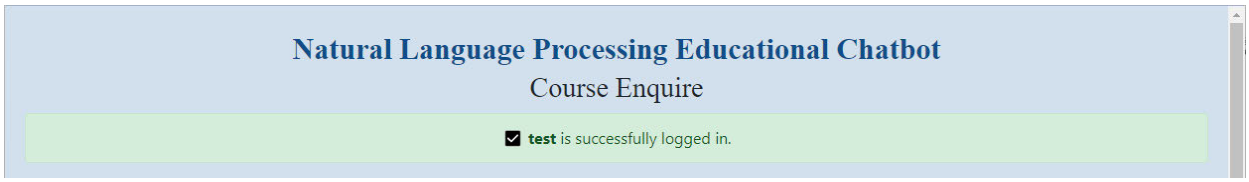


Figure 5.16 shows the header of web system

All of the web pages have a consistent layout, with the web application's name, login bar in the center of top. Also, the title of each page will be displayed under the name of the web application. The page of course registration and course update are almost identical. The consistent layout helps users to get familiar with the web application faster. Therefore, the rate of error by users and learning time could b

Movement of Eyes

Most of the content of web pages is aligned to the left, as the content will be shown swiftly in the same area. The eye will always be drawn to the left. As a result, the user's eye look will return to the identical beginning place on the following line. It provides a comfortable reading way for users, as they do not need to look around the whole page constantly.

Form fill-in

The form that appeared on the course registration and course update page has followed the form's design guidelines. Both forms have a meaningful title and give an explanatory message for each field by setting placeholders. Also, those forms allow users to press the tab keys to move the cursor to the next field, which is a convenient and straightforward mechanism for moving the cursor. Besides, the form has a logical grouping of fields; the first group is general course information, the second group is about the tutorial, the third group is about assessment, and the last group is about course syllabus information.

5.1.3 Website Evaluation Procedures

As mentioned in section 3, the website will evaluate by the below metric which defined by Ghattas et al. (2020). Yet, as the website is in local host, so some of the metric could be performed. In general, the time of loading, responding and starting are within 1 seconds. Also, there are no broken links and overwhelmed of request or size (see Table 5.1).

Metric	**Result**	**Tool(s) of checking**
Response Time	Within a second	Chrome developer tool
Load Time	Within a second	Chrome developer tool
Broken Links	No	Chrome developer tool
No.of Request	15	Chrome developer tool
Page Size	1.3Mb	Chrome developer tool
Mark up validation	Not applicable, as the website is local host	Nil
Design Optimization	Not applicable, as the website is local host	Nil
Page Speed	Within a second	Chrome developer tool
Start time render	Within a second	Chrome developer tool
Connect Time	Within a second	Chrome developer tool
DNS look up	Not applicable, as the website is local host	Nil

Table 5.1 shows the header of Website Performance Measurement Metric

The web system also passed all the test cases shown in Table 5.2. To sum up, the performance of the website is up to standard.

Test case	Pages	Procedures	Result
Check login function	Login Page	• Login with wrong Id • Login with wrong password • Login with correct id and password	Pass
All links available	All Pages	• Click all the buttons that links to another page	Pass

Result displayed according to filtering requirement	Enquiry Page	• Enter different filtering requirement and cross check the result	Pass
Check registration function	Registration Page, Duplicate Page and Update Page	• Fill in the form and press create button, then check is the result can enquire through enquiry page	Pass
Check update function	Update Page	• Modify the forms information and press update button, then check is the result can enquire through enquiry page	Pass
Check generates report function	Enquiry Page	• Press the 'generate report' button to see if an excel file will be downloaded. Then, open the file to cross check the data with database	Pass
Check Responsive Layout	All Pages	• Resize the window of browser to see whether the content is responsive	Pass
Check view record details	Duplicate page and update page	• Click 'view/duplicate' or 'view/update' button in enquiry page, then check if the record data displayed in the new page	Pass

Table 5.2 shows the test case result of web system

5.2. Virtual Assistant Performance Evaluation

5.2.1. Interface

Users can access the virtual assistant through the link https://t.me/NLPEdu_Chatbot/. When users open the link, they can press the button "SEND MESSAGE" to start using the virtual assistant.

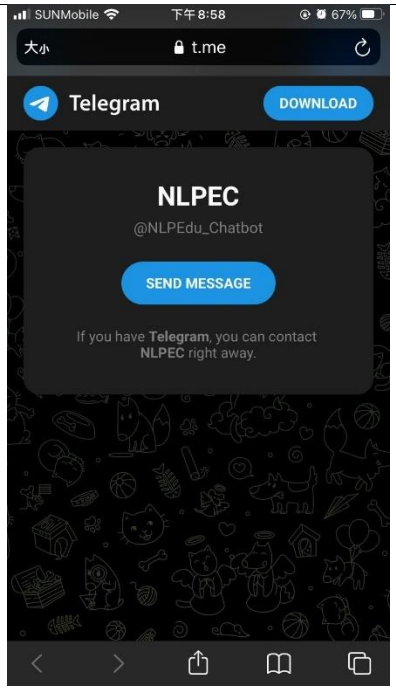
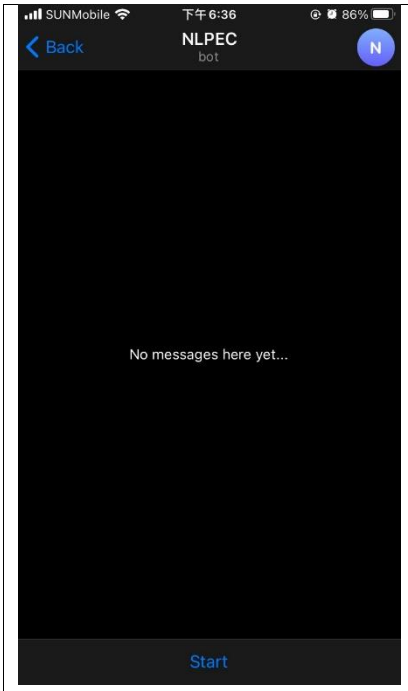
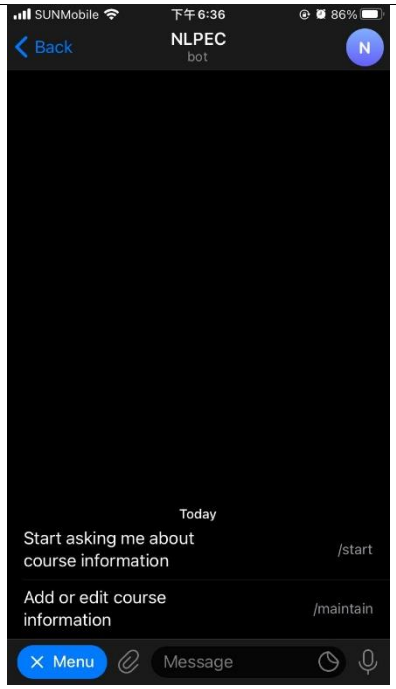


Figure 5.17 shows the access page of telegram bot

If users are using the virtual assistant first time, they will see “Start” button located in the bottom of the screen. Users can directly press “Start” to interact with virtual assistant. On the other hand, there is a blue button named “menu” placed in the left side of the text box, users can press the menu button and choose “/start” to start the conversation in anytime. When the assistant receive the message “/start”, it will require user to enter the course code that users would like to performs actions.





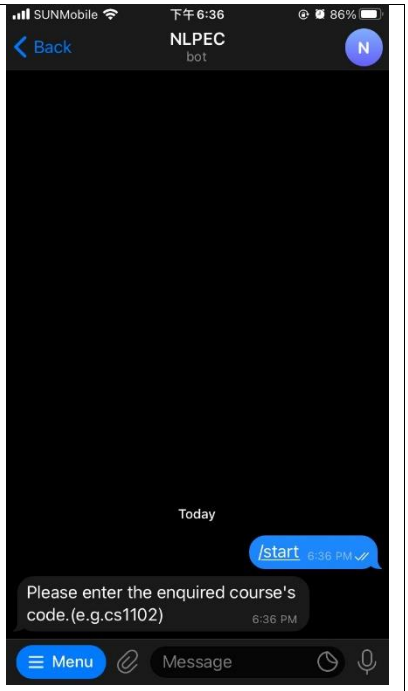


Figure 5.18 shows the initial layout of the virtual assistant	*Figure 5.19 shows how to start the conversation with the virtual assistant*	*Figure 5.20 the result of successfully start the conversation of virtual assistant*

We will input Cs1302 as an example. In the database, it could have more than one CS1302 course data with different teachers or in different semesters. So that, the assistant will query the database to find all tuples with course code CS1302, and return the semester and lecturer information for users to select a wanted course. After users input the correct ID of the course, four buttons with function name will be appear in the keyboard. Users can select one of the functions by pressing the button. After user clicked the function, user can keep interacting with the same function. If user wants to switch another functon or switch to another course, they need to input "/start" again.

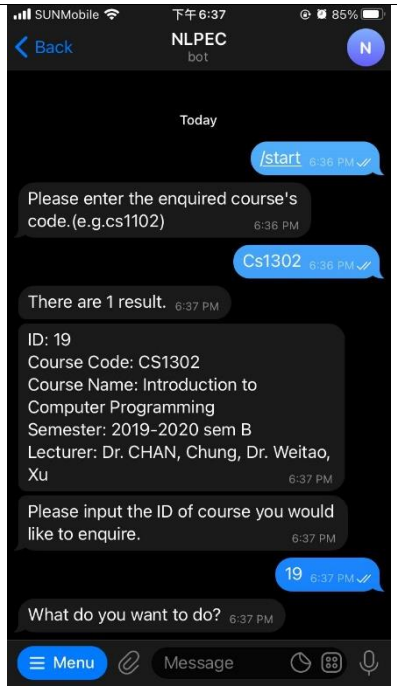


Figure 5.21 shows the progress of selecting a course

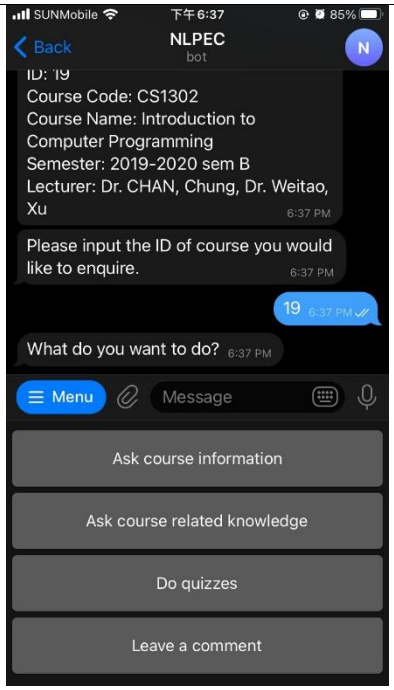


Figure 5.22 shows four functions for user to select

When users click 'Ask course information', the assistant will ask users what do they want to know. Users can directly ask questions about the course general information, such as teacher of the course and description of the course. The assistant will pass the question of users to the Course General Information Classification Model. The model will return a classification result

which correspond to the attribute of the table. Then, another function will retrieve the course information of that attribute and then return the answer to user.

Figure 5.23 shows user can ask question

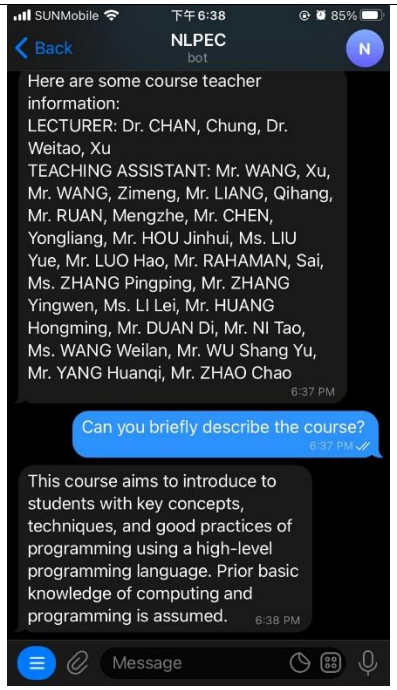
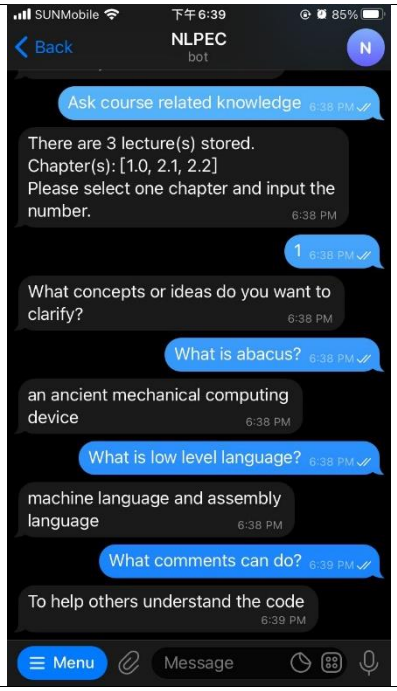


Figure 5.24 shows user can keep asking question

When users click ‘Ask course related knowledge’, the assistant will query the database to find is the lecture of this course stored in the database. If existed, the assistant will return all the chapter of lecture that is stored in the database. Users need to select one chapter and then input the number of the chapter. Each chapter of lecture are stored in the database separately because the need of the Extractive Based Question Answering model. As the model need to input context and question, the assistant need to select a most related context from the lecture according to the question. The purpose of selecting lecture is to nail down the scope of passage, so that the accuracy of selecting a correct context that contains the answer are higher.



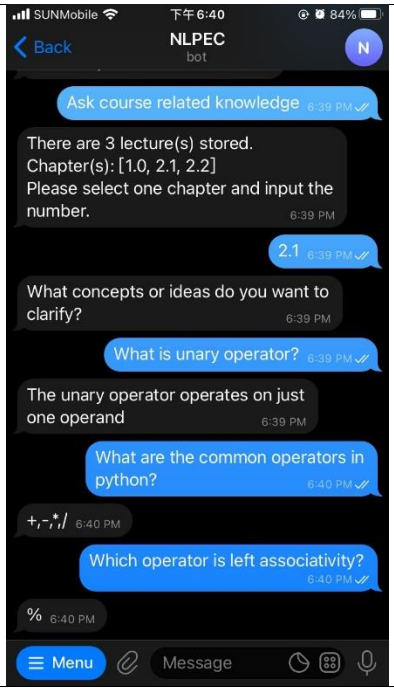
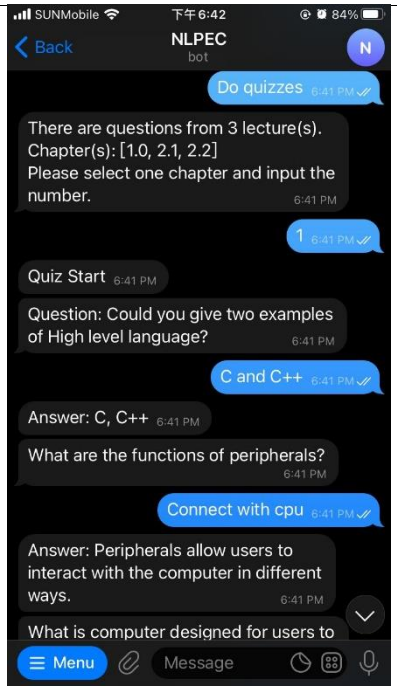


Figure 5.25 and 5.26 shows user can start asking question of lecture 1

When users click ‘Ask course related knowledge’, the assistant will first query the database to check whether this course have any question for user to do quiz. The quiz will immediately start when user inputted the number of chapter. The answer of question will be sent after user reply with his answer. The question will keep sending to user unless all the question are answered or user stop replying the assistant.



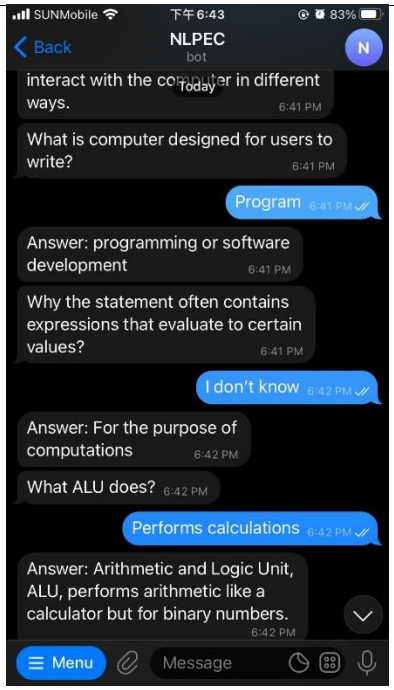


Figure 5.27 and 5.28 shows user doing quizzes

User can leave a comment of the course, then the assistant will pass the comment to the Sentiment Analysis Classification model. Then, the assistant will return the classification result to user and thanks user for commenting.

Figure 5.29 shows user leaving comment

The conversation between assistant and user in “ask course general information”, “ask course related knowledge” and “Leave a comment” function will be stored in the database. As mentioned in section 4.2, authorized users can access the web system and click generate report to view these data.

In the menu button, user can click on “/maintain” to get the link of the Web-based Course Information System.

Figure 5.30 shows how user can get the link of web system

5.2.2. Virtual Assistant Test Case Result

Test case	Procedures	Result
Check models are executables	• Tested functions of 'Ask course information', 'Ask course related knowledge' and 'Leave a comment'	Pass
Check exception handling	• Interact with assistant with unexpected responses	Pass
Check assistants send appropriate response	• Test all the functions	Pass
Saved conversation in database	• Tested functions of 'Ask course information', 'Ask course related knowledge' and 'leave a comment', then check the database	Pass

5.2.3. Course General Information Classification Model

The task of Course General Information Classification Model is to identify the category of question, so that the predicted label could use to retrieve corresponding information from database and return response to user.

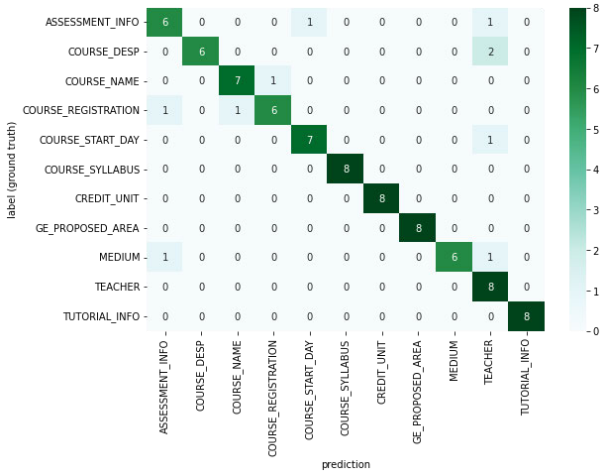
As mentioned before, CityU General Course Information Dataset have 334 samples, 246 samples are used to train the model while 88 samples are used to validate the trained model. Regarding to the small dataset, I have set the value of batch size to a relatively small value. To select a best performance model, I constructed the classification model with DistilBERT, RoBERTa and Bert-base-uncased with same training parameters, same testing and training dataset splitting, batch size of 4 and fine tune for 5 epochs over the data.

Model Version	Training Accuracy	Training Loss	Validation Accuracy	Validation Loss
distilbert-base-cased	0.975609756097561	0.5310992512010759	0.8863636363636364	0.6742283695123412
roberta-base	0.9715447154471545	0.6636751289329221	0.8977272727272727	0.6140299412337217
bert-base-uncased	0.91869918699187	1.0431174203272788	0.8068181818181819	1.0633371431719174

Table 5.3 shows the overall accuracy and loss of three model version

As shown in table 5.3, models with training accuracy more than 0.97 are distilbert-base-cased and roberta-base version. Also, they have more than 0.8 validation accuracy. Among the two models, the difference in training accuracy, training loss, validation accuracy and validation loss are less than 0.15. Yet, the bert-base-uncased version model performs the worst, all of it statistic are in the last place. As the training loss and validation loss do not have a big different, it indicates that the model are not overfitting or underfitting. The possible reason of accuracy on training and validation date are lower than the others models is the bert-base-uncased version of learning algorithum need more data than the other two models to learn.

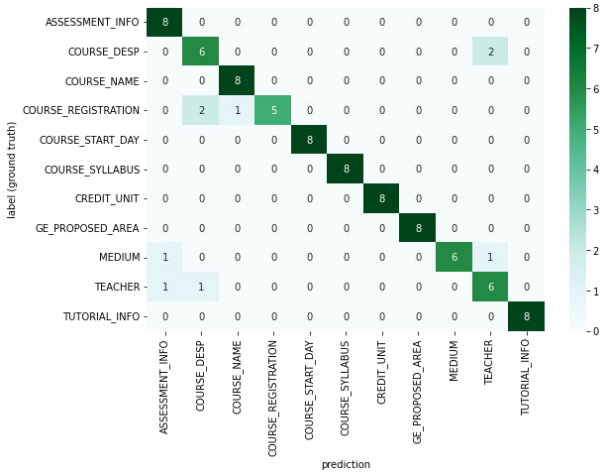
distilbert-base-cased's Confusion Matrix



distilbert-base-cased's Classification Report

	precision	recall	f1-score	support
ASSESSMENT_INFO	0.75	0.75	0.75	8
COURSE_DESP	1.00	0.75	0.86	8
COURSE_NAME	0.88	0.88	0.88	8
COURSE_REGISTRATION	0.86	0.75	0.80	8
COURSE_START_DAY	0.88	0.88	0.88	8
COURSE_SYLLABUS	1.00	1.00	1.00	8
CREDIT_UNIT	1.00	1.00	1.00	8
GE_PROPOSED_AREA	1.00	1.00	1.00	8
MEDIUM	1.00	0.75	0.86	8
TEACHER	0.62	1.00	0.76	8
TUTORIAL_INFO	1.00	1.00	1.00	8
accuracy			0.89	88
macro avg	0.91	0.89	0.89	88
weighted avg	0.91	0.89	0.89	88

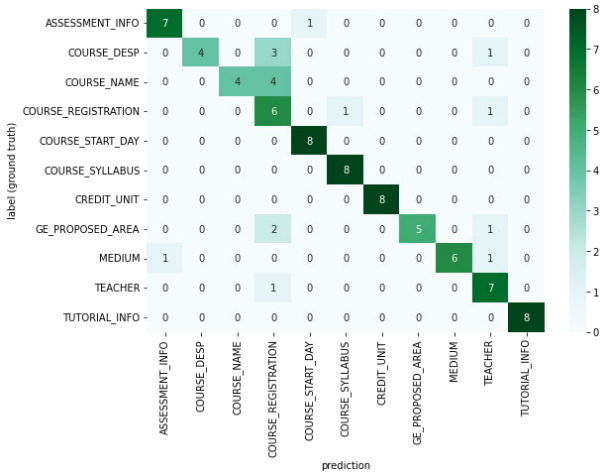
roberta-base‘s Confusion Matrix



roberta-base’s Classification report

	precision	recall	f1-score	support
ASSESSMENT_INFO	0.80	1.00	0.89	8
COURSE_DESP	0.67	0.75	0.71	8
COURSE_NAME	0.89	1.00	0.94	8
COURSE_REGISTRATION	1.00	0.62	0.77	8
COURSE_START_DAY	1.00	1.00	1.00	8
COURSE_SYLLABUS	1.00	1.00	1.00	8
CREDIT_UNIT	1.00	1.00	1.00	8
GE_PROPOSED_AREA	1.00	1.00	1.00	8
MEDIUM	1.00	0.75	0.86	8
TEACHER	0.67	0.75	0.71	8
TUTORIAL_INFO	1.00	1.00	1.00	8
accuracy			0.90	88
macro avg	0.91	0.90	0.90	88
weighted avg	0.91	0.90	0.90	88

bert-base-uncased’s Confusion Matrix



bert-base-uncased’s Classification report

	precision	recall	f1-score	support
ASSESSMENT_INFO	0.88	0.88	0.88	8
COURSE_DESP	1.00	0.50	0.67	8
COURSE_NAME	1.00	0.50	0.67	8
COURSE_REGISTRATION	0.38	0.75	0.50	8
COURSE_START_DAY	0.89	1.00	0.94	8
COURSE_SYLLABUS	0.89	1.00	0.94	8
CREDIT_UNIT	1.00	1.00	1.00	8
GE_PROPOSED_AREA	1.00	0.62	0.77	8
MEDIUM	1.00	0.75	0.86	8
TEACHER	0.64	0.88	0.74	8
TUTORIAL_INFO	1.00	1.00	1.00	8
accuracy			0.81	88
macro avg	0.88	0.81	0.81	88
weighted avg	0.88	0.81	0.81	88

Table 5.4 shows the confusion matrix and classification report of models

Table 5.4 shows a more detail result of the classification of the three models. Firstly, all models predict all the labels of course syllabus, credit unit and teacher correct completely. The predictions performance of roberta-base and distilbert version shows in confusion matrix are very similar, the difference of correct prediction numbers is only 1. Roberta-base version predicts 79 data correctly while distil bert version predicts 78 data correctly. Yet, in the classification report, the precision of roberta-base version are mostly higher than the precision of distilbert version, it means the roberta-base version have generally higher accuracy of positive predictions. So, the roberta-base version has the best performance in the validation data.

On the other hand, the bert-base-uncased version have difficulty on classifying course name and course description. When compared to the other two models, the bert-base-uncased version have higher generalization error which can’t accurately classify records it has never seen before. The bert-base-uncased version model might not suitable for small dataset.

5.2.3. Course Related Knowledge Question Answering Model

I have tried to apply two different datasets to train the QA model. At first, the Python Course Question Answering Dataset which contains 556 samples in training set and 110 samples in validation set is used on DistilBERT based model. I trained the model with 16 batch size and 3 epochs. However, the performance is not up to standard.

DistilBERT based model: Total training_loss=2.94686437536169

Epoch	Training Loss	Validation Loss
1	No log	2.896615
2	No log	2.199396
3	No log	2.076346

Table 5.5

Generally, the training loss and validation loss are > 2. Also, the total training loss is higher than the validation loss of last epoch, it shows the model have some overfitting.

Then, I use the trained model to do prediction on validation dataset. Some of the result is selected and shown below.

Record	1	2
Question	What is Abacus?	How can we make the code more readable?
Expected Answer	an ancient mechanical computing device	use the format function to perform string formatting
Predicted Answer	1st : [{'score': 5.239053, 'text': 'computer can perform arithmetic calculations, it is not only a calculator that is bigger and more advanced.'}, 11th :{'score': 3.7713544, 'text': 'Abacus is an ancient mechanical computing device.'}	1st: [{'score': 7.4859166, 'text': 'The'} 8th: {'score': 4.2628155, 'text': 'To make the code more readable, we can use the format function to perform string formatting.'}

Table 5.6 shown two prediction result of validation set

I have stored the best 20 answers for each prediction in a list. In record 1, the predicted answer with highest score 5.239053 is not the expected answer. The correct answer is the eleventh item

with score 3.7713544. In record 2, the predicted answer with highest score of 7.4859166 is not the expected answer. The closest similar answer is the eightieth item with score of 4.2628155. Then, I have increased the epoch from 3 to 10 to train again. Yet, it didn't improve the performance of the model. The prediction result is very similar to the previous attempt. These poor performances could be due to not enough data.

After that, I decided to apply squad dataset, a large dataset for training and use the Python Course Question Answering Dataset for validation. I trained the distilbert-base-cased, roberta-base and bert-base-uncased versions of models with the same training parameters, which are 16 batch sizes and 3 epochs.

<table>
<tr><th></th><th>Training Details</th><th>Metric evaluation</th></tr>
<tr><td>Distilbert</td><td><table><tr><th>Epoch</th><th>Training Loss</th><th>Validation Loss</th></tr><tr><td>1</td><td>1.225000</td><td>1.151014</td></tr><tr><td>2</td><td>0.949400</td><td>1.122609</td></tr><tr><td>3</td><td>0.770100</td><td>1.153468</td></tr></table>'train_loss': 1.0918292205257152</td><td>{'exact_match': 76.76442762535478, 'f1': 85.06019032289329}</td></tr>
<tr><td>Roberta</td><td><table><tr><th>Epoch</th><th>Training Loss</th><th>Validation Loss</th></tr><tr><td>1</td><td>0.898500</td><td>0.862591</td></tr><tr><td>2</td><td>0.688500</td><td>0.857835</td></tr><tr><td>3</td><td>0.532800</td><td>0.898151</td></tr></table>'train_loss': 0.77037415660645</td><td>{'exact_match': 86.13055818353831, 'f1': 92.36150687340479}</td></tr>
<tr><td>Bert Base Uncased</td><td><table><tr><th>Epoch</th><th>Training Loss</th><th>Validation Loss</th></tr><tr><td>1</td><td>1.064100</td><td>1.012542</td></tr></table></td><td>{'exact_match': 80.80416272469253, 'f1': 88.34214928698543}</td></tr>
</table>

	2	0.775300	1.022953	
	3	0.601100	1.068614	
	'train_loss': 0.9056592008580472			

Table 5.7 shows training statistics of question answering model

Among the three models, the roberta-base version model performs the best. Its have highest exact match and 1f1 score. Also, this model have the lowest taining loss and validation loss among the thre models.

5.2.4. Student Feedback Classification Model

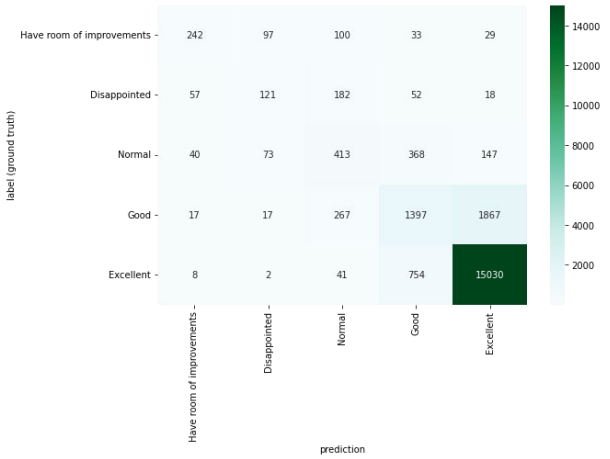
The Student Feedback Classification Model aims to classify the sentiment from a comment. We will use it to classify the student feedback for the teacher to evaluate the class arrangement. This model used the 100K Coursera's Course Reviews Dataset. There are 85483 samples in the training set and 21372 samples in the validation set. I trained the distilbert-base-cased, roberta-base and bert-base-uncased version of models with the same training parameters, batch size 4 and 3 epochs.

Model Version	Training Accuracy	Training Loss	Validation Accuracy	Validation Loss
distilbert-base-cased	0.839610723810415	0.4194910049426973	0.8049316863185476	0.5275995742542965
roberta-base	0.8273171758761053	0.4501360671342703	0.8158805914280367	0.5230050924234397
bert-base-uncased	0.849693538576709	0.3966001360280544	0.8023114355231144	0.591341947003735

Table 5.8 shows overall accuracy and loss

As shown in table 5.9, all models have training accuracy over 0.82 and validation accuracy over 0.8. While looking at the training loss and validation loss, the distilbert-base-cased version has a 0.108 difference, the roberta-base version has a 0.073 difference, and the bert-base-uncased version has a 0.2 difference between the two losses. As the roberta-base version has the most minor difference, it is the most perfect fitting model among all models.

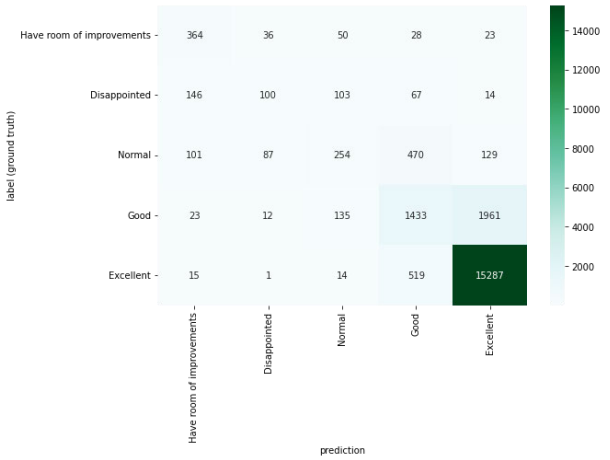
distilbert-base-cased's Confusion Matrix



distilbert-base-cased's Classification Report

	precision	recall	f1-score	support
Have room of improvements	0.66	0.48	0.56	501
Disappointed	0.39	0.28	0.33	430
Normal	0.41	0.40	0.40	1041
Good	0.54	0.39	0.45	3565
Excellent	0.88	0.95	0.91	15835
accuracy			0.80	21372
macro avg	0.58	0.50	0.53	21372
weighted avg	0.78	0.80	0.79	21372

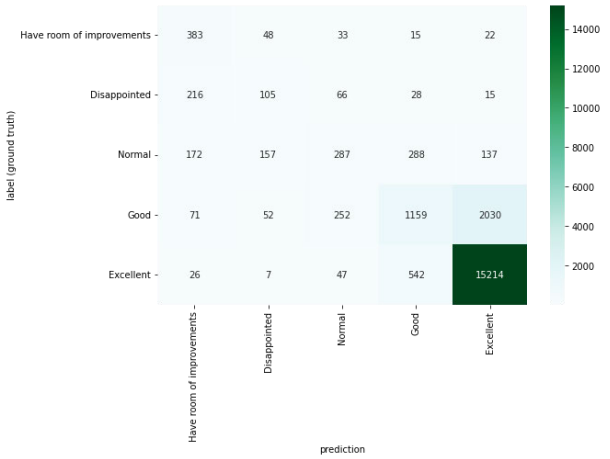
roberta-base's Confusion Matrix



roberta-base's Classification report

	precision	recall	f1-score	support
Have room of improvements	0.56	0.73	0.63	501
Disappointed	0.42	0.23	0.30	430
Normal	0.46	0.24	0.32	1041
Good	0.57	0.40	0.47	3564
Excellent	0.88	0.97	0.92	15836
accuracy			0.82	21372
macro avg	0.58	0.51	0.53	21372
weighted avg	0.79	0.82	0.80	21372

bert-base-uncased's Confusion Matrix



bert-base-uncased's Classification report

	precision	recall	f1-score	support
Have room of improvements	0.44	0.76	0.56	501
Disappointed	0.28	0.24	0.26	430
Normal	0.42	0.28	0.33	1041
Good	0.57	0.33	0.41	3564
Excellent	0.87	0.96	0.92	15836
accuracy			0.80	21372
macro avg	0.52	0.51	0.50	21372
weighted avg	0.78	0.80	0.78	21372

Table 5.9 shows the confusion matrix and classification report of models

Table 5.9 shows that the data of each label in the validation set are not evenly distributed. Firstly, all models are capable of classifying the excellent label. Secondly, the roberta-base and bert-base-uncased versions can correctly classify the label "have room of improvement". However, all models have a different level of difficulty to classify the labels "Disappointed", "Normal" and "Good". The distilbert-base-cased is mostly classified wrongly between two labels. Yet, the

roberta-base and bert-base-uncased versions have tremendous difficulty classifying these labels. The predictions are evenly predicted to five labels. The performance of distilbert-base-cased and roberta-base is very similar, yet the roberta-base has generally higher precision than the other two models. So, the roberta-base version model is the best model among the three models.

count label 1 (have room of improvement)	2468
count label 2 (Disappointed)	2249
count label 3 (Normal)	5067
count label 4 (Good)	18037
count label 5 (Excellent)	79044

Table 5.10 shows the number of label in the dataset

Label	Comment
2(Disappointed)	good course
	Great and free.
	Nice, but very basic.
	Too easy, very basic
3 (Normal)	assignment is a little bit messy.
	Very confusing course for those who have no previous experience in Java.
	Good to build python basic data the concept clear
4 (Good)	Very easy and easy to understand
	So far, so good! Just about to start week 2 of this course and it has been good so far.
5 (Excellent)	Very Intresting
	Excellent introduction to accesing and manipulating data from the web or other sources. Light load of lectures and easy assignments it can be completed in a few hours and allows you to wrestle further complexities on your own.

Table 5.11 shows some sampling in the dataset

The dataset is the main reason that all models do not have outstanding performance. I counted the number of each label and found that the number of labels was not even. Label ‘good’ and

‘excellent’ have more than 10000 samples, yet the other three labels have less than 5100 samples. Without enough training data, the model could not perform well in predictions.

Instead of the uneven number of labels, the comment quality is also affected. For example, some students think the course is ‘good’, ‘great’, ‘nice’ but only give 2 marks. Some students believe the course is ‘very confusing’, ‘good’, ‘messy’ but only give 3 marks. These strong emotional words could hardly affect the model to learn the label ‘normal’ feature, so the predictions are wrong. Also, some words appear in different labels, such as ‘easy’, some give 2 marks, some give 4 marks, and some give 5 marks. The comment that rates 4 and 5 are very similar explains why three models will misclassify the ‘good’ label to the ‘excellent’ label.

6. Conclusion

6.1 Critical Review

The most uncomplicated aspect of finishing this project was developing the website and chatbot; nevertheless, the most challenging aspect was training the model and preparing the data.

Since I learned how to build a website and use telegram to build a chatbot in previous classes and internships, I am familiar with the debug and testing process. Yet, this project needs to develop two platforms. There will be a lot of trivial parts that need to be handled. For example, both platforms also need to validate the data entered by the user, so a lot of validation functions need to be written. In order not to miss any function and error, I wrote test cases and tested each unit after one stage was completed.

Throughout the model training process, as I followed the tutorial's instructions for building the model, all I needed to do was comprehend what the tutorial is doing and apply it to my model. The unexpected difficulty encountered while training the model results from Colab's limited resources. Certain models require a significant amount of time to train on large data sets. During that time, the colab may disconnect. Eventually, in order to prevent having to retrain the model, I learned to store checkpoints so that even if the model training is interrupted, I can resume training using checkpoints.

The most challenging aspect is gathering data sets. The quality of the dataset could affect the performance of models. For instance, the Course General Information Classification Model and Question Answering Model perform better than the sentiment analysis model. That is because the first model is used my constructed dataset, I collected data from several linked websites and constructed a qualified dataset. Another model uses the SQUAD dataset, a famous and great dataset. However, the student review data set that was downloaded for the sentiment analysis model is not up to standard:

1. The data set contains garbled characters and non-English words. Finding and rearranging all inapplicable data in a data set of more than 100,000 records take time.
2. The number of labels is not even in the dataset.
3. The quality of the comment is not consistent.

Therefore, selecting a proper and qualified dataset is of paramount importance.

6.2 Summary of Achievement

To begin, this project is a new approach to developing a virtual assistant. The virtual assistant is linked to an online platform that enables non-administrators to manage course materials and registration. The advantage is that the majority of work can be accomplished without the involvement of administrators.

Additionally, this project has resulted in the development of a completely functional virtual assistant since all platforms and functionalities have been completed. Anyone interested in using it is allowed to clone the code and connect it to their own database. He can then begin utilizing this virtual helper.

Furthermore, this project develops three distinct types of models for various objectives. In each type of model, three different pre-trained models were utilized to fine-tune the dataset. By comparing different datasets and analyzing their performance, it is helpful to quantify which model is best for a small dataset and which is ideal for a large dataset.

Meanwhile, the web system's interface is meticulously created in accordance with user studies and design standards. As a result, the online system could give an excellent user experience to a wide variety of users. Lastly, the interface and function were assessed using the seven stages of action models, which attempted to minimize the gulf of execution and evaluation.

6.3 Suggestions for Extension

6.3.1 Connect with CityU database and canvas

To let virtual assistants help answer student's questions, the database to which the virtual assistants are connected needs to be constantly updated. To solve this problem, we have created a website for this project. Yet, with this approach, teachers need to enter the information into the website manually. If the teacher has more than one course in a semester, they need to input many materials for each course manually. This process of inputting materials takes time. Suppose an API can directly request information from the CityU database and Canvas. In that case, the course materials will be automatically updated every day, which can save a lot of time and avoid wrong typing information.

6.3.2 Lecture Rephase

The current question-answer model requires a plain text file of lecture notes. In this project, I manually rewrote three chapters. Some of the contents in the notes are in point form. It is

necessary to convert these texts into complete sentences. To avoid this process, a new NLP model could be created to try on rewriting non-complete sentences to complete sentences.

6.3.3 question generation to make the quiz

The quizzes provided by the current quiz function are to think by myself and then store the questions and answers in the database. However, when I created these materials, I found that it took time to think about the questions and answers. A question generation model could be developed to speed up the progress so that multiple-choice or short questions could be generated by inputting a piece of text. The model could save time and be capable of making a large number of quizzes for students to review by themselves.

7. Appendices

7.1. Monthly Log

Month	Monthly Log
October 2021	For documentation, I reviewed the comment of the project plan from the supervisor and finished the draft of interim report, including the introduction and literature review. For the website, all of the functions and pages were constructed. They are pending for tests. For the virtual assistant, I connected the telegram API and the database. Ph.D. student, Shu Qi, has given me some information about fine-tuning the Bert model. I am studying from it and trying to build the model of the first function.
November 2021	The interim report I, which includes an introduction, literature review, preliminary design, evaluation design, and implementation schedule, was completed. Also, the web application passed the unit test and system test. The first model had tried to construct, yet still pending tests.
December 2021	I fine-tuned the multi-class classification model of the "Ask course information" function using the "bert-base-uncased" model of Huggingface's transformer. The model was successfully embedded and run in the backend of the virtual course assistant. The assistant gave out corresponding responses according to the given questions. Afterward, I constructed other models with "DistilBERT" to compare their performances. I will perform more tests on those models and select the best one. After meeting with the supervisor, I decided to narrow down the scope of the "Ask course-related knowledge" and "Quiz generation" to a python course CS1302. I collected the related teaching materials and used them to construct a dataset of "ask course-related knowledge." For interim report II, I started to write the result of the "Ask course information" models in the "Preliminary result" section.

January 2022	• Constructed and compared 2 models of the Multi-class classification with different pre-trained models • Prepared dataset of extractive based question answering model and finetuned the first model using a distilling BERT base • I had a meeting with P.hD student ShuQi to review current progress and confirmed the details included in the fyp report • Submitted the Interim report II
February 2022	• Constructed and compared in total 5 models of the Multi-class classification with different pre-trained models • Embedded the best model of Multi-class classification into backend • Constructed 1 extractive based question answering model with distilling BERT base • Constructing the sentiment analysis model
March 2002	• Completed the final report and all the codings

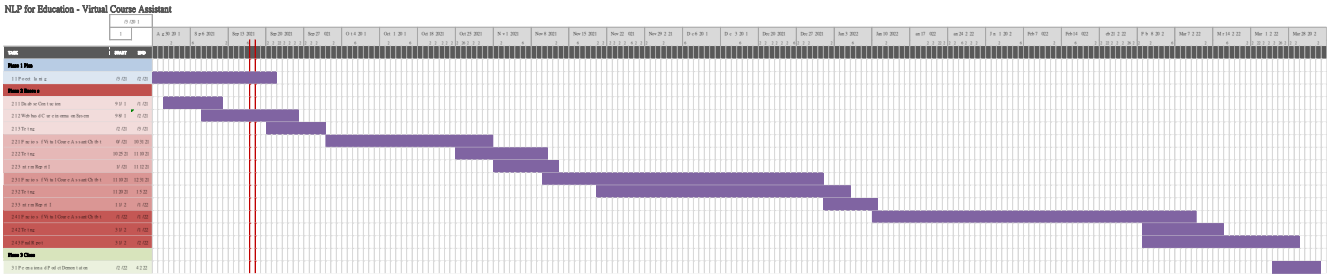
7.2. Plan and Schedule for Implementation and Testing

7.2.1. High-level Project Schedule

Stage	Description	Deliverables	Start Date	End Date
Stage 1 – Project Planning				
1	Project kick-off	1. Project Plan	30 Aug 2021	21 Sep 2021
Stage 2 – Database Construction and Web-based Course information System				
2.1	Database Construction	1. Database Design Diagrams 2. Data Collection	1 Sep 2021	11 Sep 2021
2.2	Web-based Course information System	Functions: 1. User Sign Up 2. User Login 3. Course Registration 4. Course Update 5. Course Enquiry	8 Sep 2021	25 Sep 2021
2.3	Testing	1. Unit and System Testing Result 2. Software modules passing the tests	20 Sep 2021	30 Sep 2021
Stage 3 –Functions of Virtual Course Assistant Chatbot, Improvements to Stage 2 deliverables and Interim report I				
3.1	Functions of Virtual Course Assistant Chatbot	Functions: 1. Answer Course General Information	1 Oct 2021	31 Oct 2021
3.2	Testing	1. Unit Testing Result 2. New software modules passing the tests	25 Oct 2021	10 Nov 2021
3.3	System Documentation	Interim Report I	1 Nov 2021	12 Nov 2021
Stage 4 – Functions of Virtual Course Assistant Chatbot , Improvements to Stage 3 deliverables and Interim Report 2				
4.1	Functions of Virtual Course Assistant Chatbot	Functions: 1. Answer Course Related Knowledge Concepts 2. Provide Quizzes	12 Nov 2021	20 Feb 2021
4.2	Testing	1. Unit Testing Result 2. New software modules passing the tests	21 Feb 2021	28 Feb 2022

Stage	Description	Deliverables	Start Date	End Date
4.3	System Documentation Update	Interim Report II	15 Jan 2022	30 Jan 2022
Stage 5 –Functions of Virtual Course Assistant Chatbot, Improvements to Stage 4 deliverables and Final Report				
5.1	Functions of Virtual Course Assistant Chatbot	1. Generate report with student Feedback and sentiment analysis to Teachers	1 March 2022	5 March 2022
5.2	Testing	1. System and System Integration Testing Result 2. New software modules passing the test	6 March 2022	20 March 2022
5.2	System Documentation Update	Final Report	15 March 2022	29 March 2022
Stage 6 – Presentation and Product Demonstration				
6.1	Presentation and Product Demonstration	Presentation Slides	25 March 2022	2 Apr, 2022

7.2.2. Gantt Chart



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